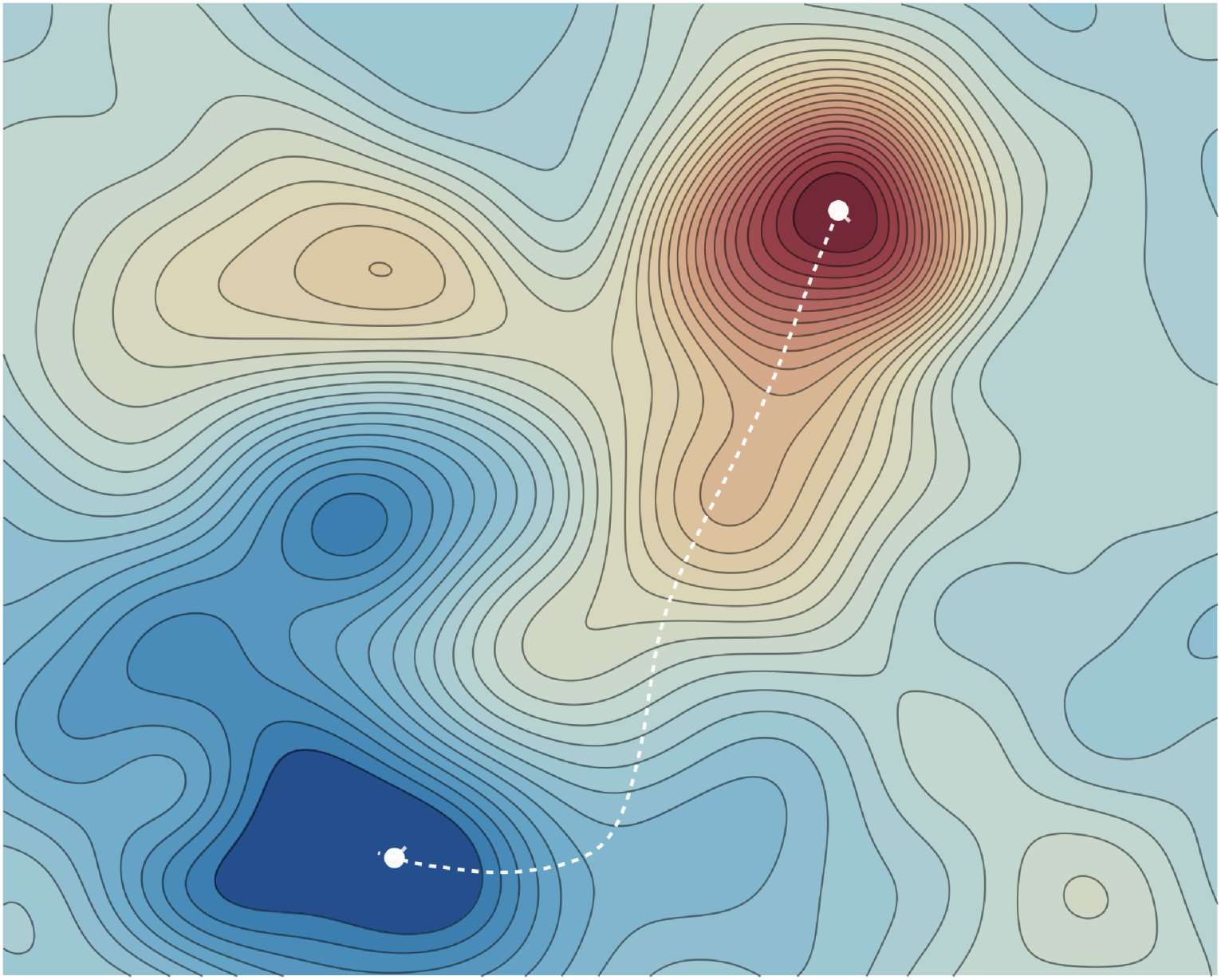

Machine Learning

Author: Shaoheng Yan (PhotonYan)

Instructor: Prof. Muhan Zhang

Institute: Institute of Artificial Intelligence, Peking University

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Chapter 1 Linear Regression

Introduction

- *ERM*
- *Gradient Descent*
- *Ridge Regression (L_2 regularization)*
- *Lasso Regression (L_1 regularization)*

1.1 Basic Knowledge

1.1.1 Linear Regression

Settings.

- Dataset: $D = \{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. Here, y_i denotes the regression target, while x_i represents the input features used to predict y_i.
- Linear Model: $f(x) = w^\top x + b$, with weight $w \in \mathbb{R}^d$ and bias $b \in \mathbb{R}$.
- Linear Model: $f(x) = w^\top x + b$, with weight $w \in \mathbb{R}^d$ and bias $b \in \mathbb{R}$.

Note

This definition is equivalent to an inner product: $\hat{y} = w^\top x + b$. $\hat{y} = w^\top x + b$.

Definition 1.1 (Learnable / Trainable Parameters)

Learnable parameters are those that can be updated during the training process. ♣

Quiz. How to determine whether a parameter is learnable? How to determine w and b?

Ans: **ERM** (Empirical Risk Minimization)

- Loss function. Squared Loss (SE) is commonly used during optimization. The training objective can be written as:

$$\operatorname*{argmin}_{w,b} \frac{1}{n} \sum_{i\in[n]} \left(y_i - (w^\top x_i + b)\right)^2 \operatorname*{argmin}_{w,b} \frac{1}{n} \sum_{i\in[n]} \left(y_i - (w^\top x_i + b)\right)^2 \tag{1.1}$$

  The blue factor $1/n$ can be omitted in theoretical analysis, but is often kept in practice to stabilize the loss function during implementation.

Quiz: How to optimize the parameters?

Ans: **Gradient Descent** (as a traditional ML method). In the case of linear regression:

$$\frac{\partial \mathcal{L}}{\partial b} = -2 \sum_{i\in[n]} (y_i - w^\top x_i - b) \tag{1.2}$$

$$\frac{\partial \mathcal{L}}{\partial w} = -2 \sum_{i\in[n]} (y_i - w^\top x_i - b) x_i \tag{1.3}$$

Note

*In the field of machine learning, the gradient of a scalar with respect to a vector is itself a vector **(not a covector)**.*

This means:

$$\frac{\partial \mathcal{L}}{\partial w} = \begin{pmatrix} \frac{\partial \mathcal{L}}{\partial w_1} \\ \frac{\partial \mathcal{L}}{\partial w_2} \\ \vdots \\ \frac{\partial \mathcal{L}}{\partial w_d} \end{pmatrix} = \left(\frac{\partial \mathcal{L}}{\partial w_1}, \frac{\partial \mathcal{L}}{\partial w_2}, \cdots, \frac{\partial \mathcal{L}}{\partial w_d} \right)^\top \tag{1.4}$$

See the definition of matix derivatives (assuming no special structure in the matrix) in Matrix Cookbook Chapter 2.

Note

Here are some commonly used derivative formulas:

$$\frac{\partial x^\top x}{\partial x} = 2x \tag{1.5}$$

$$\frac{\partial a^\top x}{\partial x} = a, \quad \frac{\partial Ax}{\partial x} = A^\top \tag{1.6}$$

$$\frac{\partial x^\top Ax}{\partial x} = (A + A^\top)x \tag{1.7}$$

Remark

Both sides of an equation must have the same dimension. This principle can be used as a consistency check.

We optimize the parameters by subtracting a scalar multiple of the gradient from the parameters, considering the physical meaning of the gradient: the direction of the steepest **increase**.

Definition 1.2 (Hyperparameter)

A parameter that is fixed during optimization and specified before the training process. ♣

That is:

$$w' = w - \alpha \frac{\partial \mathcal{L}}{\partial w}, \quad b' = b - \alpha \frac{\partial \mathcal{L}}{\partial b} \tag{1.8}$$

Optimization will stop when the norm of the parameter update becomes smaller than a given hyperparameter.

1.2 Closed-Form of Linear Regression

Proposition 1.1

*Linear Regression has **Closed-Form** solution.* ♡

Settings.

- Matrix $X_0 := (x_1, \cdots, x_n)^\top$;
- Matrix $X := (X_0, \mathbb{1}) \in \mathbb{R}^{n \times (d+1)}$;
- $y = (y_1, \cdots, y_n)^\top \in \mathbb{R}^n$;
- $\hat{w} = (w^\top, b)^\top \in \mathbb{R}^{d+1}$.

Then the loss function of $\hat{w}$ can be written as:

$$\mathcal{L}(\hat{w}) = (y - X\hat{w})^\top (y - X\hat{w}) = \|y - X\hat{w}\|_2^2 \tag{1.9}$$

Here, $\|\cdot\|_p$ denotes the *p-norm* of a vector.

Note

Vectors can sometimes be treated as scalars, since linearity ensures that the validity of a proposition can be extended to any finite dimension.

Notice that the optimization stops when $\partial\mathcal{L}(\hat{w})/\partial\hat{w} = 0$. Under this condition, the parameters can be solved from the above constraint by following steps:

$$\frac{\partial\mathcal{L}(\hat{w})}{\partial\hat{w}} = -2X^\top(y - X\hat{w}) \tag{1.10}$$

Note

Both dimensional analysis and calculation using Leibniz's rule lead to the same result as the formula above:

$$\begin{aligned}\mathcal{L}(\hat{w}) &= y^\top y - 2y^\top X\hat{w} + \hat{w}^\top X^\top X\hat{w}\\ \partial_{\hat{w}}\mathcal{L}(\hat{w}) &= -2X^\top y + 2X^\top X\hat{w}\\ &= -2X^\top(y - X\hat{w})\end{aligned}$$

Remark

More matix formulas are available in Matrix Cookbook.

Thus, the target of the optimization satisfied:

$$X^\top y = X^\top X\hat{w} \tag{1.11}$$

That is:

$$\hat{w} = (X^\top X)^{-1}X^\top y \tag{1.12}$$

when $X^\top X$ invertible (non-singular / full-rank).

Example 1.1

When does $X^\top X$ not invertible?

Solution

$X \in \mathbb{R}^{n\times(d+1)}$:

- $d + 1 > n$. *Brief Proof:* $\operatorname{rank}(X^\top X) = \operatorname{rank}(X) \le \min(n, d+1) = n < d + 1$.
- *X has repeated columns. Proof is trivial.*

When $X^\top X$ isn't invertible:

1. If $\operatorname{rank}(X^\top X, X^\top y) > \operatorname{rank}(X^\top X)$, $\hat{w}$ has no solution;
2. $\hat{w}$ has infinity solution o.w.

Situation 1 is **impossible** because both $X^\top X$ and $X^\top y$ can be represented in the column space of $X^\top$. Therefore, the optimization problem must have a solution, which may be either unique or infinite.

As an infinite set of solutions makes it difficult to determine which estimate of $\hat{w}$ to choose, we apply L_2 **regularization** to linear regression, which is commonly referred to as **Ridge Regression**. That is:

$$\mathcal{L}_{L_2} := \mathcal{L}(\hat{w}) + \boxed{\lambda\|\hat{w}\|_2^2}, \tag{1.13}$$

where $\lambda > 0$ is a hyperparameter. Notice that $\|\hat{w}\|_2^2 = \sum_{i=1}^{d+1}\hat{w}_i^2$, L_2 regularization prevents any single dimension from being assigned an excessively large weight, and encourages the model to make use of more dimensions during training.

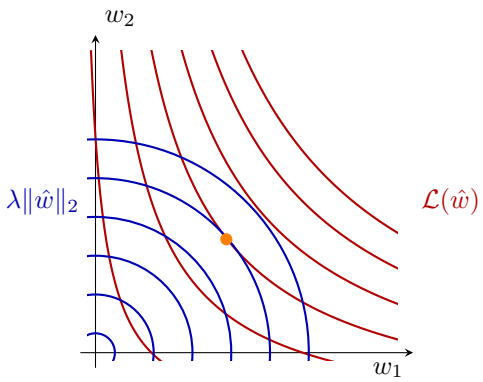


Figure 1.1: Illustration of L_2 regularization. The contours represent level sets of the regularized loss $\mathcal{L}(\hat{w}) + \lambda\|\hat{w}\|_2^2$, which take the form of concentric ellipses (circle in the plot).

During ridge regression, we minimize the $\mathcal{L}_{L_2}$:

$$\underset{\hat{w}}{\operatorname{argmin}}\,(y - X\hat{w})^\top(y - X\hat{w}) + \lambda\hat{w}^\top\hat{w} \tag{1.14}$$

The optimization stops when:

$$\frac{\partial \mathcal{L}_{L_2}}{\partial \hat{w}} = -2X^\top y + 2X^\top X\hat{w} + 2\lambda I\hat{w} = 0 \tag{1.15}$$

$$\Rightarrow (X^\top X + \lambda I)\hat{w} = X^\top y \tag{1.16}$$

Proposition 1.2

$X^\top X + \lambda I$ always invertible. ♡

Proof

Since $X^\top X$ is a real symmetric matrix, we have the eigen-decomposition $X^\top X = U\Lambda U^\top$, where $\Lambda = \operatorname{diag}(\lambda_1, \cdots, \lambda_{d+1})$. Moreover, as $X^\top X \succeq 0$ (positive semi-definite), it follows that $\forall i \in [d+1]$, $\lambda_i \geq 0$. Note that:

$$\lambda I = \lambda U U^\top \tag{1.17}$$

since U is an orthogonal matrix. Hence:

$$X^\top X + \lambda I = U(\Lambda + \lambda I)U^\top \tag{1.18}$$

For all $i \in [d+1]$, we have:

$$\lambda_i + \lambda > \lambda_i \geq 0 \tag{1.19}$$

Thus, $X^\top X + \lambda I$ is a full-rank matrix. ■

Remark

Numerical issues may still occur even if $X^\top X$ is full rank (e.g., when eigenvalues λ_k are close to zero). The L_2 regularization factor λ mitigates this issue by shifting the eigenvalues upward, thereby improving numerical stability during training.

Another regularization method often used is L_1 regularization, where the loss function is defined as:

$$\mathcal{L}_{L_1} := \mathcal{L}(\hat{w}) + \boxed{\lambda\|\hat{w}\|_1} \tag{1.20}$$

L_1 regularization can induce sparsity in $\hat{w}$, which works in contrast to L_2 regularization. Specifically, L_1 regularization encourages the model to rely on only a small subset of input features, effectively performing **feature selection**.

Linear regression with L_1 regularization is called **Lasso Regression** (Least Absolute Shrinkage and Selection

Operator).

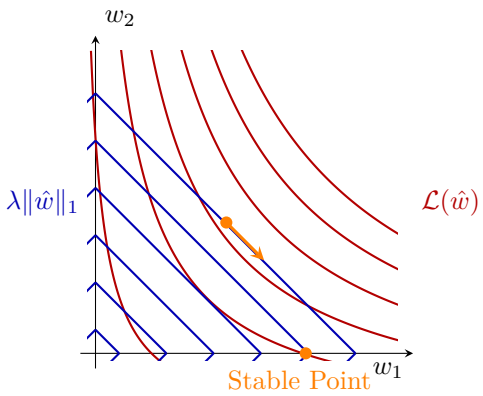


Figure 1.2: Illustration of L_1 regularization. The contours represent level sets of the regularized loss $\mathcal{L}(\hat{w}) + \lambda\|\hat{w}\|_1$, which take the form of nested diamonds (squares rotated by 45° in the plot).

1.3 Geomeric View of LR

Ideally, we would like to solve $X\hat{w} = y$. If y lies on the hypersurface

$$\mathcal{M}(X) := \mathrm{Span}(X) = \{Xw : w \in \mathbb{R}^{d+1}\} \subset \mathbb{R}^n \tag{1.21}$$

then the equation admits an exact solution. In most cases, however, $y \notin \mathcal{M}(X)$, so no exact solution exists. Nevertheless, we can always find an estimator $\hat{w}$ such that $\mathcal{P}_{\mathcal{M}(X)}y = X\hat{w}$, where $\mathcal{P}_{\mathcal{M}(X)}$ denotes the orthogonal projection onto the hypersurface $\mathcal{M}(X)$.

Proposition 1.3

$$\hat{y} = X\hat{w} \quad \Rightarrow \quad \hat{w} \textit{ is solution to LR.} \tag{1.22}$$

Proof

$$\begin{aligned} y - \hat{y} \perp \mathcal{M}(X) \quad &\Rightarrow \quad y - X\hat{w} \perp \mathcal{M}(X) \\ &\Rightarrow \quad X^\top(y - X\hat{w}) = 0 \quad \Rightarrow \quad \hat{w} = (X^\top X)^{-1}X^\top y \end{aligned}$$

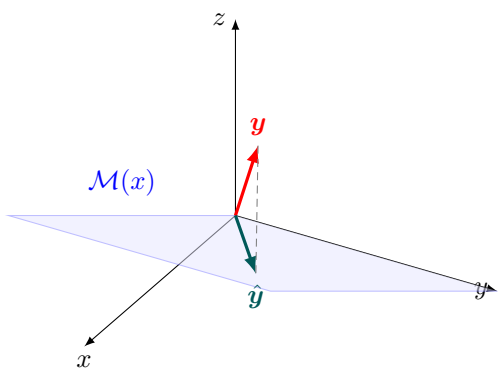


Figure 1.3: Orthogonal projection interpretation of linear regression. The predicted vector $X\hat{w}$ is obtained as the projection of y onto the hypersurface $\mathcal{M}(X) = \{Xw : w \in \mathbb{R}^{d+1}\}$, which is a linear subspace in the classical case.

Chapter 2 Logistic Regression

Introduction

- *Binary Classification Problem*
- *Cross Entropy*
- *Sigmoid Regression*
- *Maximum a posteriori*

2.1 Classification

Example 2.1

Binary Classification Problem

Settings.

- Dataset: $D = \{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. Here, y_i denotes the classification target, while x_i represents the input features used to predict y_i.
- Model in Logistic Regression: In logistic regression, we start with a linear model $f(x) = w^\top x + b$. Unlike in ordinary regression, where the target variable lies in $\mathbb{R}$, here the target space collapses to $\{0, 1\}$. Thus, we need a function that maps real-valued outputs into this discrete set. Moreover, in many applications it is desirable to obtain not only a hard classification decision (0 or 1), but also a *soft* prediction: the probability of each class. Such a probabilistic interpretation provides both the likelihood estimate and the corresponding classification outcome.

Can we directly use a linear model to fit $p(y = 1 \mid x = x_i)$, as we did in the previous chapter? The answer is *no*. This is because there is a mismatch between the range of a linear model output (which lies in $\mathbb{R}$) and the valid domain of probabilities, $[0, 1]$.

To resolve this issue, we introduce a transformation function called the **sigmoid** function. The sigmoid maps any real-valued input into the interval $[0, 1]$, making it suitable for modeling probabilities. It is defined as:

Definition 2.1 (Sigmoid Function)

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{2.1}$$

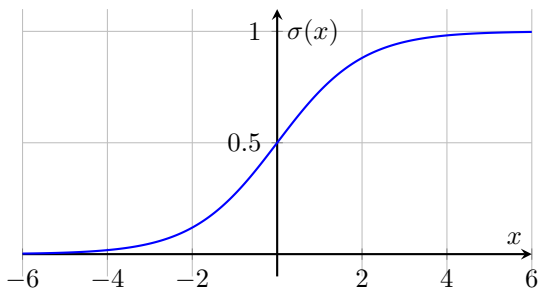


Figure 2.1: The sigmoid function $\sigma(z)$ over the interval $z \in [-6, 6]$.

The sigmoid function enjoys several elegant properties.

Theorem 2.1

$$1 - \sigma(z) = \sigma(-z)$$

Proof

This follows directly from the definition of $\sigma(z)$, or equivalently, by observing the symmetry of its graph. ■

To convert soft prediction results into binary outputs $\{0, 1\}$, we introduce a threshold: when $\sigma(z) = 0.5$, the model makes a hard prediction.

Definition 2.2 (Separating Hyperplane)

The condition $\sigma(z) = 0.5$ defines the separating hyperplane. *It partitions the input space $\mathbb{R}^d$ into two regions, thereby transforming probabilistic predictions into binary classification outcomes.* ♣

The normal vector w is perpendicular to this hyperplane and points towards the region where the model predicts class 1 (i.e., where $p(y = 1 \mid x) > 0.5$). We ensure this property by choosing the orientation of w accordingly.

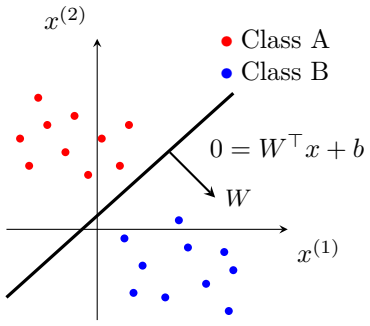


Figure 2.2: Classification by heperplane.

Model definition is clear, and now we turn our attention to parameter optimization. The question is: how can we find the proper w, b that achieve the best performance? The key problem here is to identify a suitable loss function that can be optimized via gradient descent.

Notice that:

$$P(y = 1 \mid x = x_i) = \sigma(f(x_i)) = \frac{1}{1 + \exp(-w^\top x + b)}. \tag{2.2}$$

We introduce a new method rather than continuing with ERM, by using **Maximum Likelihood Estimation (MLE)**. MLE is naturally designed to address probability modeling problems.

Definition 2.3 (Maximum Likelihood Estimation (MLE))

MLE aims to find parameters such that the likelihood of $P(y = y_i \mid x = x_i)$ is maximized. ♣

Remark

For brevity, we write $P(y = y_i \mid x = x_i)$ as $P(y_i \mid x_i)$.

Definition 2.4 (Likelihood)

The likelihood on the entire training data is defined as

$$\prod_{i \in [n]} P(y_i \mid x_i; w, b) \tag{2.3}$$

assuming the samples are independent. ♣

According to the discussion above, in logistic regression we :

$$P(y_i \mid x_i) = \begin{cases} \sigma(w^\top x_i + b), & y_i = 1, \\ 1 - \sigma(w^\top x_i + b), & y_i = 0. \end{cases} \tag{2.4}$$

Therefore, the likelihood function can be expanded as:

$$\prod_{i \in [n]} \sigma(w^\top x_i + b)^{y_i} \left(1 - \sigma(w^\top x_i + b)\right)^{1-y_i} \tag{2.5}$$

The above form is intuitive once we recall that $x^0 = 1$.

To ensure floating-point precision, given the large amount of data and the monotonicity of the logarithm function, we transform the likelihood into the maximization of the log-likelihood:

$$\operatorname*{argmax}_{w,b} \sum_{i \in [n]} \left[y_i \log \sigma(w^\top x_i + b) + (1 - y_i) \log \left(1 - \sigma(w^\top x_i + b)\right) \right] \tag{2.6}$$

This is the final objective in MLE.

MLE can be transformed into ERM by applying argmin to the negative log-likelihood. We define the **Cross-entropy Loss**:

$$\mathcal{L}(w, b) := - \sum_{i \in [n]} \left[y_i \log \sigma(w^\top x_i + b) + (1 - y_i) \log \left(1 - \sigma(w^\top x_i + b)\right) \right] \tag{2.7}$$

Thus, minimizing the cross-entropy loss is equivalent to maximizing the log-likelihood, making the equivalence between MLE and ERM immediate

Note

Why is the above loss called the Cross-entropy loss? The name originates from information theory. Entropy is defined as:

$$H(P) = \sum_y P(y) \log \frac{1}{P(y)} = - \sum_y P(y) \log P(y). \tag{2.8}$$

In other words, rarer events (with smaller probability) carry more information, and entropy measures the expected amount of information. In our context, we can evaluate the information content of the prediction for $y = \hat{y}_i$ given $x = x_i$ as:

$$\begin{aligned} H(P) &= - \sum_{\hat{y}_i \in \{0,1\}} P(y = \hat{y}_i \mid x_i) \log P(y = \hat{y}_i \mid x_i) \\ &= -[P(y = 1 \mid x_i) \log P(y = 1 \mid x_i) + P(y = 0 \mid x_i) \log P(y = 0 \mid x_i)] \end{aligned} \tag{2.9}$$

Notice that under our estimation, $P(y = 1 \mid x_i) = \sigma(f(x_i; w, b))$ and $P(y = 0 \mid x_i) = 1 - \sigma(f(x_i; w, b))$. In practice, we substitute the empirical distribution of samples for the true distribution when comparing the negative log-likelihood with entropy. This is why the terminology of entropy from information theory is carried over to name this loss term.

Note

The formal definition of cross entropy between two propbability distributions q and p $H(q, p)$ is defined by:

$$H(q, p) = - \sum_y q(y) \log p(y) \tag{2.10}$$

Definition 2.5 (KL-Divergence)

The KL-Divergence between two distributions p and q is defined by:

$$\mathrm{KL}(q \| p) = \sum_y q(y) \log \frac{q(y)}{p(y)} \tag{2.11}$$

♣

KL-Divergence measures the difference between two given distributions. In particular, $\mathrm{KL}(q\|p)$ differs from the cross-entropy by only a constant term $H(q)$:

$$\mathrm{KL}(q\|p) = H(q,p) - H(q). \tag{2.12}$$

In other words, KL-Divergence quantifies the extra number of bits required when we use p to approximate the ground-truth distribution q.

Back to the main content. Recall that a closed-form solution can be derived for the linear regression problem, as mentioned in the previous chapter. Here, we would like to investigate whether logistic regression also admits a closed-form solution.

Following the same steps we applied in linear case, we define $\hat{x} = (x^\top, 1)^\top \in \mathbb{R}^{d+1}$ and $\hat{w} = (w^\top, b)^\top \in \mathbb{R}^{d+1}$, thus $f(x) = \hat{w}^\top \hat{x}$:

$$\begin{aligned}
\mathcal{L}(\hat{w}) &= -\sum_{i\in[n]} y_i \log \sigma(\hat{w}\hat{x}_i) + (1-y_i)\log(1-\sigma(\hat{w}^\top\hat{x}_i)) \\
&= -\sum_{i\in[n]} y_i \log \frac{1+\exp(\hat{w}^\top\hat{x}_i)}{1+\exp(-\hat{w}^\top\hat{x}_i)} - \log(1+\exp(\hat{w}^\top\hat{x}_i)) \\
&= -\sum_{i\in[n]} y_i(\hat{w}^\top\hat{x}_i) - \log(1+\exp(\hat{w}^\top\hat{x}_i)) \qquad (2.13)
\end{aligned}$$

Take the derivative:

$$\frac{\partial\mathcal{L}(\hat{w})}{\partial\hat{w}} = -\sum_{i\in[n]}\left[y_i\hat{x}_i - \frac{\exp(\hat{w}^\top\hat{x}_i)}{1+\exp(\hat{w}^\top\hat{x}_i)}\hat{x}_i\right] \tag{2.14}$$

$$= -\sum_{i\in[n]}\left[y_i - P(y=1\mid x_i)\right]\hat{x}_i \tag{2.15}$$

By gradient descent, the parameter is updated as $\hat{w} \leftarrow \hat{w} + \alpha\sum_{i\in[n]}\Big(y_i - P(y_i=1\mid x_i)\Big)\hat{x}_i$. This makes sense, since the term $y_i - P(y_i=1\mid x_i)$ directly measures the prediction error on sample i, and the update moves $\hat{w}$ a small step along the direction of the input $\hat{x}_i$ to reduce this error.

If for all i, we have $y_i = P(y=1\mid x_i)$, then the model predicts every label y_i perfectly. At this point, optimization reaches a stationary solution. If the training data admits such a perfect solution:

Definition 2.6 (linearly separable)

If all points can be separated by a linear model without error, we say the dataset is linearly separable. ♣

Example **??** is linearly separable, and the final state leads to $\|W\| \to \infty$, $\|b\| \to \infty$. However, this situation is not desirable in practice, since it implies poor robustness. Hence a natural question arises: under the condition of linear separability, how can we find a well-chosen separating hyperplane that maximizes robustness? The answer will be presented in the next chapter, where we introduce the Support Vector Machine (SVM). The SVM optimizes $\hat{w}, \hat{b}$ by maximizing the margin (the distance between data points and the separating hyperplane), instead of simply minimizing the cross-entropy loss.

Although logistic regression may suffer from divergence of parameters under separable data, it often achieves better performance than SVM in practice, due to the following reasons:

1. In most real-world problems, the data are not linearly separable;
2. Applying L_2 regularization can effectively prevent parameter divergence.

Remark

Why can't we use squared loss for classification? The reason is that in classification tasks such as logistic regression, the label $y \in \{0,1\}$ should be interpreted as a categorical outcome rather than a numerical quantity.

Example 2.2

Multi-Class Classificaiton (Softmax Regression)

We can combine k sub-classifiers to solve a k-class classification task. Specifically, we apply a sigmoid-like transformation to each sub-linear model $f_k(x) = w_k^\top x + b_k$, and obtain the probability of class k using a normalized expression:

$$P(y = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_{j=1}^{k} \exp(w_j^\top x + b_j)}. \tag{2.16}$$

Note

*There is a close analogy between **softmax regression** in machine learning and the **partition function** in statistical physics.*
In softmax regression, the probability of assigning an input x to class k is

$$P(y = k \mid x) = \frac{\exp(\theta_k^\top x)}{\sum_{j=1}^{K} \exp(\theta_j^\top x)}. \tag{2.17}$$

Here the denominator

$$Z(x) = \sum_{j=1}^{K} \exp(\theta_j^\top x) \tag{2.18}$$

serves as a normalizing constant*, ensuring that the probabilities over all classes sum to* 1.
In statistical physics, for a system with possible states s of energy $E(s)$, the probability of observing state s under the Boltzmann distribution is

$$P(s) = \frac{\exp(-\beta E(s))}{Z} = -\frac{1}{\beta}\frac{\partial \log Z}{\partial E(s)}, \quad Z = \sum_{s} \exp(-\beta E(s)), \tag{2.19}$$

where Z is the partition function*. It normalizes the distribution and encodes all thermodynamic properties of the system.*
Thus, the role of $Z(x)$ in softmax regression is mathematically analogous to the role of the partition function Z in statistical physics: both are log-sum-exp normalizers that transform unnormalized scores (energies or logits) into proper probability distributions.

The softmax regression method has several advantages:

1. It is unified and normalized: $\sum_{j\in[j]} P(y = k \mid x) = 1$;
2. The amplify effect of exp function: when $f_k(x) \gg f_j(x), \forall j \neq k$, then $P(y = k \mid x) = 1$.

We apply the same MLE procedure to the multi-class classification task, which leads to the following optimization problem:

$$\underset{\{w_k, b_k\}}{\operatorname{argmax}} \sum_{i\in[n]} \log \frac{\exp(w_k^\top x_i + b_k)}{\sum_{j=1}^{k} \exp(w_j^\top x_i + b_j)} \tag{2.20}$$

Note

In modern neural networks, softmax regression is widely used as the standard classification method.

Remark

When $k = 2$, softmax regression reduces to logistic regression via the reparameterization $w = w_1 - w_2$ and $b = b_1 - b_2$.

2.2 Rethink of Linear Regression

Example 2.3

MLE Explanation for Linear Regression

Definition 2.7 (Gaussian/Normal Distribution)

$$x \sim \mathcal{N}(\mu, \sigma^2) \quad \Leftrightarrow \quad P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{2.21}$$

♣

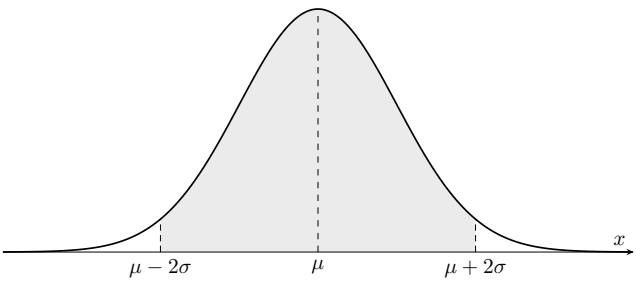


Figure 2.3: Normal distribution (95%).

While the Central Limit Theorem (CLT) does not imply that most datasets are normally distributed, it motivates modeling additive noise as Gaussian. We assume:

$$y = \underbrace{w^\top x + b}_{\text{latent model}} + \underbrace{\varepsilon}_{\text{noise}}, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \tag{2.22}$$

where σ^2 is a hyperparameter characterizing the noise scale. Then:

$$P(y \mid x; w, b, \sigma^2) = \mathcal{N}\big(y \,\big|\, w^\top x + b,\ \sigma^2\big) \tag{2.23}$$

The log-likelihood takes the form:

$$\begin{aligned} \operatorname*{argmax}_{w,b} \sum_{i\in[n]} \log \mathcal{N}\big(y_i \,\big|\, w^\top x_i + b,\ \sigma^2\big) &= \operatorname*{argmax}_{w,b} \sum_{i\in[n]} \left[-\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{\big(y_i - (w^\top x_i + b)\big)^2}{2\sigma^2}\right] \\ &\Leftrightarrow \operatorname*{argmin}_{w,b} \sum_{i\in[n]} \big(y_i - (w^\top x_i + b)\big)^2, \end{aligned} \tag{2.24}$$

where the equivalence follows by dropping constants and positive scalings. Equation (2.24) recovers ERM with the squared-loss objective.

Example 2.4

Maximum a Posteriori (MAP)

In the MLE perspective, w and b are treated as unknown fixed constants. In the Bayesian framework, however, even w, b are considered as random variables (R.V.). A "fixed constant" can be seen as a random variable with an extremely sharp distribution (near δ-distribution).

Suppose we place a Gaussian prior $P(\hat{w}) = \mathcal{N}(\hat{w} \mid 0,\ \sigma_w^2 \mathbb{I})$. The likelihood is given by

$$P(y \mid x, \hat{w};\ \sigma^2, \sigma_w^2) = \mathcal{N}\big(y \mid \hat{w}^\top \hat{x},\ \sigma^2\big) \tag{2.25}$$

We want to compute the posterior distribution of $\hat{w}$. By Bayes' rule:

$$P(\hat{w} \mid y, x) = \frac{P(y \mid x, \hat{w})\, P(\hat{w})}{P(y \mid x)}$$
$$\propto \left(\prod_{i \in [n]} P(y_i \mid x_i, \hat{w}) \right) P(\hat{w})$$

Expanding this expression:

$$P(\hat{w} \mid y, x) = \frac{1}{Z} \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i \in [n]} \left(y_i - \hat{w}^\top \hat{x}_i \right)^2 \right) \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^{d+1} \exp\left(-\frac{\hat{w}^\top \hat{w}}{2\sigma_w^2} \right)$$

Taking the negative logarithm of the posterior, note that only the terms involving $\hat{w}$ are subject to optimization:

$$-\log P(\hat{w} \mid y, x) = \sum_{i \in [n]} \left(y_i - \hat{w}^\top \hat{x}_i \right)^2 + \frac{\sigma^2}{\sigma_w^2} \|\hat{w}\|^2 + \text{Const.}$$

Letting $\lambda = \sigma^2 / \sigma_w^2$, the MAP estimator is obtained by:

$$\operatorname*{argmin}_{\hat{w}} \sum_{i \in [n]} \left(y_i - \hat{w}^\top \hat{x}_i \right)^2 + \lambda \|\hat{w}\|^2, \tag{2.26}$$

which is exactly ridge regression. Therefore, the internal consistency of the theory is demonstrated.

Chapter 3 Support Vector Machine

Introduction

- *Constrained optimization*
- *Inequality optimization*
- *KKT condition*
- *Support Vector Machine*

Support Vector Machine (SVM) is also a binary classification model, often introduced as an alternative to logistic regression, since—as we have discussed in the previous chapter—logistic regression has several notable limitations. SVM provides an elegant framework that allows us to transfer powerful results and intuition from optimization theory into machine learning.

3.1 Constrained Optimization

Unlike the unconstrained optimization problems we studied earlier, the support vector machine is originally formulated as a **constrained optimization problem**. This distinction is fundamental. In mathematics, unconstrained optimization is defined as:

$$\underset{x}{\operatorname{argmin}}\, f(x) \tag{3.1}$$

When we deal with constrained optimization, we add an auxiliary constraint $h(x) = 0$ (where both $f(x)$ and $h(x)$ are differentiable). Considering the geometric meaning of the gradient, we observe:

1. For any point on the surface defined by $h(x)$, the gradient $\nabla h(x)$ must be orthogonal to the surface.
2. For any critical point x^* that achieves a local minimum, the gradient $\nabla f(x^*)$ must also be orthogonal to the surface.

Note

The second property can be understood by analogy with electrostatics: near a conductor, the electric field must be orthogonal to the surface. Otherwise, there would be a component of the field (gradient) along the constrained sub-hypersurface, violating equilibrium.
Extending this analogy: in the electrostatic case, such a tangential component would induce currents along the conductor's surface. In the optimization setting, a tangential component corresponds to a gradient lying along the constraint manifold, implying that the objective function can still be further optimized without violating the constraint.
Moreover, the charge distribution problem in electrostatics can itself be interpreted as an energy minimization process: the charges rearrange to minimize the system's electrostatic energy. In this sense, the equilibrium state of charges on a conductor can be seen as an optimization problem, which strengthens the analogy with constrained optimization in mathematics.

Combining the two conditions above, we obtain the **necessary** condition for a local minimum point, as shown in Figure 3.1:

$$\exists \lambda \quad \text{s.t.} \quad \nabla f(x^*) + \lambda \nabla h(x^*) = 0 \tag{3.2}$$

Remark

In some contour examples, the constraint condition does not cover the entire space. In such cases, the above equation may not *always be satisfied.*

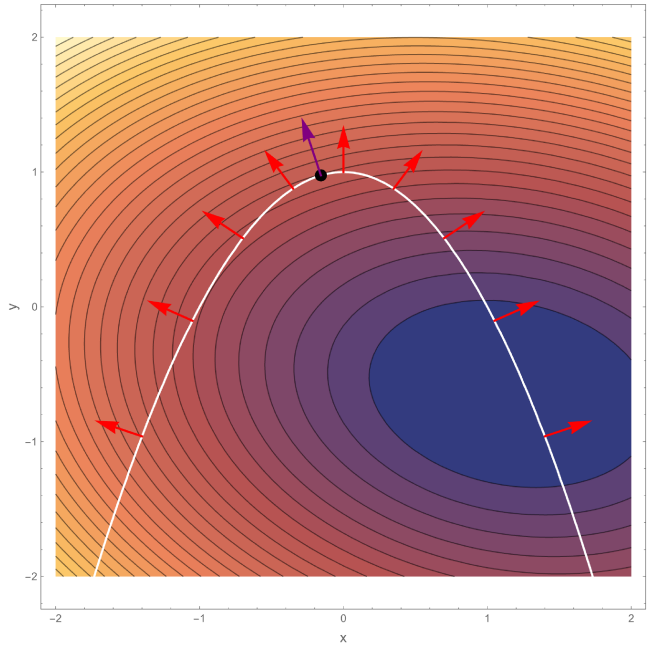


Figure 3.1: Illustration of the necessary condition for constrained local minima.

Now we introduce the *Lagrangian function*, which plays a central role in optimization theory:

$$L(x, \lambda) := f(x) + \underbrace{\lambda}_{\text{Lagrange Multiplier}} h(x) \tag{3.3}$$

Extending to the case of multiple constraints, the Lagrangian is defined as:

$$L(x, \lambda) := f(x) + \sum_{i=1}^{k} \lambda_i h_i(x) \tag{3.4}$$

Theorem 3.1

If x^* is a local minimum point, then there exists a set of multipliers $\lambda = (\lambda_1, \ldots, \lambda_k)$ such that

$$\begin{cases} \nabla_x L(x^*, \lambda) = 0, \\ \nabla_\lambda L(x^*, \lambda) = 0. \end{cases} \tag{3.5}$$

The first condition corresponds to the stationarity (minimum) condition, while the second condition enforces the constraints.

♡

Corollary 3.1

In multi-constrained problems, the minimum condition becomes

$$\nabla f(x^*) + \sum_{i=1}^{k} \lambda_i \nabla h_i(x^*) = 0. \tag{3.6}$$

♡

Remark

In higher dimensions, the constrained submanifold may possess certain free variables that can be chosen arbitrarily while still satisfying the minimum condition. Specifically, if $\nabla f(x^*)$ cannot be expressed as a linear combination of $\nabla h_1(x^*)$ and $\nabla h_2(x^*)$, it is possible to continue optimizing along the intersection submanifold defined by $h_1 = 0$ and $h_2 = 0$, as shown in Figure 3.2. Therefore, it must be emphasized that the above equation provides only a ***necessary condition****, not a sufficient one.*

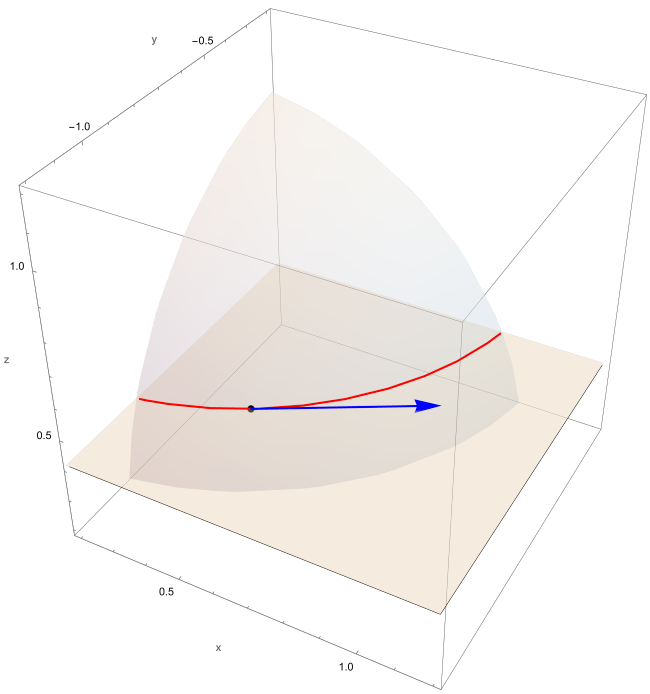


Figure 3.2: Illustration of optimization along the intersection of two constraint manifolds.

Example 3.1

Consider the following constrained optimization problem:

$$\underset{x_1,x_2}{\text{argmin}}\ x_1 + x_2 \tag{3.7}$$

subject to the constraint

$$(x_1 - 1)^2 + x_2^2 - 1 = 0 \tag{3.8}$$

Solution

$1 - \sqrt{2}$.

Note

The Lagrange multiplier method can provide critical points, but these do not always correspond to the minimum solution.

Note

In convex settings, any local minimum is also a global minimum.

A new problem arises: what if we have an inequality constraint instead of a strict equality, i.e., $g(x) \leq 0$?

3.2 Inequality optimization

We assume the feasible set of x is compact; otherwise, the existence of a minimum may be problematic.

By gradient analysis, we obtain:

1. For any point on the surface $g(x) = 0$, $\nabla g(x)$ is orthogonal to the surface and points outward from the feasible

region $g(x) \leq 0$. This is straightforward, since $\nabla g(x)$ indicates the steepest direction of increase.

2. For a local minimum point x^*:
   (a). If x^* lies on the surface $g(x) = 0$ (in other words, the constraint is *active*), then $-\nabla f(x^*)$ must be aligned with $\nabla g(x^*)$. Equivalently, $\exists\, \mu > 0$ such that
   $$\nabla f(x^*) + \mu \nabla g(x^*) = 0$$
   (b). If x^* does not lie on the surface, i.e., $g(x) < 0$ (the constraint is *inactive*), we simply require $\nabla f(x^*) = 0$. In the unified form above, this corresponds to $\exists\, \mu = 0$ such that
   $$\nabla f(x^*) + \mu \nabla g(x^*) = 0$$

Note

How should we understand the meaning of an "active" constraint? When the inequality constraint is not tight (i.e., $g(x) < 0$), it does not affect gradient descent optimization. In this case, the constraint is effectively absent, which is why we call it inactive.

Again, we define the Lagrangian as

$$L(x, \mu) = f(x) + \mu g(x) \tag{3.9}$$

where $g(x)$ denotes the inequality constraint function. Following the same steps as in the equality-constrained case, we obtain:

Theorem 3.2

If x^* is a local minimum point, then

$$\begin{cases} \nabla_x L(x^*, \mu) = 0, \\ g(x^*) \leq 0, \\ \mu \geq 0, \\ \mu \cdot g(x^*) = 0. \end{cases} \tag{3.10}$$

The last condition (the *complementary slackness*) unifies the boundary and interior cases, since either $g(x^*) = 0$ (active constraint) or $\mu = 0$ (inactive constraint) must hold.

3.3 General Case

In the most general case, we consider the optimization problem

$$\operatorname*{argmin}_x f(x) \tag{3.11}$$

subject to equality constraints $h_i(x) = 0,\ i \in [k]$, and inequality constraints $g_j(x) \leq 0,\ j \in [l]$.

The corresponding Lagrangian is defined as

$$L(x, \lambda, \mu) := f(x) + \sum_{i \in [k]} \lambda_i h_i(x) + \sum_{j \in [l]} \mu_j g_j(x) \tag{3.12}$$

In this setting, a local minimum x^* must satisfy a series of conditions known as the **Karush–Kuhn–Tucker (KKT) conditions**:

Theorem 3.3 (KKT Conditions)

If x^* is a local minimum point, then there exist multipliers (λ, μ) such that

$$\begin{cases} \nabla_x L(x^*, \lambda, \mu) = 0, & \\ h_i(x^*) = 0, & \forall i \in [k], \\ g_j(x^*) \leq 0, & \forall j \in [l], \\ \mu_j \geq 0, & \forall j \in [l], \\ \mu_j \cdot g_j(x^*) = 0, & \forall j \in [l]. \end{cases} \tag{3.13}$$

♡

In most cases, the above conditions cannot be solved explicitly in closed form.

3.4 Support Vector Machine

3.4.1 Hard Margin

Consider a linearly separable dataset $\{(x_i, y_i)\}$ with $y_i \in \{-1, 1\}$, and a linear model

$$f(x) = w^\top x + b \tag{3.14}$$

Although introducing a regularization term can make the solution unique by adjusting the hyperparameter λ, this does not resolve the issue of determining which solution $f(x; \lambda)$ is preferable among all possible choices, since λ is itself arbitrary.

Our goal is for the model to generalize well to test data drawn from the same distribution as the training set. A direct and intuitive way to achieve this is to choose the separating hyperplane that maximizes the distance between the plane and the nearest datapoints.

Definition 3.1 (Support Vectors)

The datapoints that lie closest to the separating hyperplane, such that the vectors orthogonal to the hyperplane connect the hyperplane to these datapoints, are called support vectors.

♣

In the case of SVM, we aim to find the **Max-Margin Classifier**, i.e., the hyperplane that maximizes the minimum margin across all training points. Here, the *margin* refers to the distance from a point to the hyperplane, which is equivalently the norm of the vector associated with its support vector.

Remark

Why does $w^\top x + b = 0$ represent a hyperplane? Consider the hyperplane through the origin, given by $w^\top x = 0$. Translating this plane until it passes through some point $x_0 \notin \{x \mid w^\top x = 0\}$, we obtain

$$w^\top (x - x_0) = 0$$

Setting $-w^\top x_0 = b$, we recover the general hyperplane equation

$$w^\top x + b = 0$$

Remark

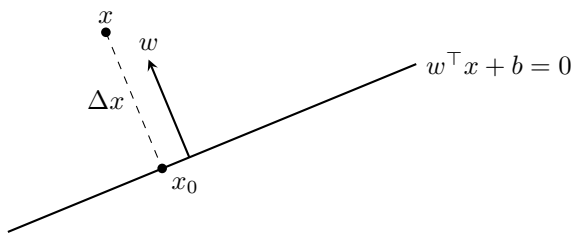
Note that the interpretation of b differs between the two cases, $y = w^\top x + b_f$ and $0 = w^\top x + b_e$. We denote b_f as the bias in the functional form and b_e as the bias in the implicit equation form to distinguish between the two. In the latter case, we observe that $|b_e| = \|w\|\|x_0\|$, which represents the offset determined by the reference point x_0.

The distance between a datapoint and the hyperplane is given by:

Theorem 3.4

$$d = \frac{|w^\top x + b|}{\|w\|}. \tag{3.15}$$

Geometrically, we obtain:



Proof

1. *Consider*

$$\begin{cases} x_0 + \Delta x \dfrac{w}{\|w\|} = x, \\ w^\top x_0 + b = 0, \end{cases}$$

so that

$$\begin{aligned} & w^\top x_0 + w^\top \Delta x \frac{w}{\|w\|} = w^\top x \\ \Rightarrow \quad & -b + w^\top \Delta x \frac{w}{\|w\|} = w^\top x \\ \Rightarrow \quad & \Delta x \|w\| = w^\top x + b \end{aligned}$$

This matches exactly the expression we obtained in (3.15).

2. *From the Lagrangian perspective:*

$$\operatorname*{argmin}_{x_0} \frac{1}{2}\|x - x_0\|^2 \quad s.t.\ w^\top x_0 + b \quad \Rightarrow \quad L(x_0, \lambda) = \frac{1}{2}\|x - x_0\|^2 + \lambda(w^\top x_0 + b)$$

$$\Rightarrow \quad \begin{cases} \nabla_{x_0} L(x_0, \lambda) & = 0 \quad \Rightarrow \quad x_0 - x + \lambda w = 0 \\ \nabla_\lambda L(x_0, \lambda) & = 0 \quad \Rightarrow \quad w^\top x_0 + b = 0 \end{cases}$$

This leads to the same solution as above.

■

When $\Delta := (w^\top x + b)/\|w\| > 0$, the point x lies on the positive side of the hyperplane; when $\Delta = 0$, x lies exactly on the hyperplane; and when $\Delta < 0$, x lies on the negative side.

We define

$$\gamma_i = \frac{y_i(w^\top x_i + b)}{\|w\|} \tag{3.16}$$

to incorporate both the label information and the margin (distance to the hyperplane). If $\gamma_i > 0$, the point x_i is correctly classified.

Define the margin

$$\gamma = \operatorname*{argmin}_{i \in [n]} \gamma_i \tag{3.17}$$

and the SVM optimization problem can be formalized as

$$\operatorname*{argmax}_{w,b} \gamma, \quad \text{s.t. } \forall i, \ \frac{y_i(w^\top x_i + b)}{\|w\|} \geq \gamma \tag{3.18}$$

This formulation is problematic, since γ is not an independent variable but is defined as the minimum of the γ_i's. To

resolve this, suppose (x_0, y_0) is a point achieving the minimum margin γ. Then

$$\gamma = \frac{y_0(w^\top x_0 + b)}{\|w\|} \tag{3.19}$$

Thus, the SVM objective can be reformulated as

$$\underset{w,b}{\operatorname{argmax}} \frac{y_0(w^\top x_0 + b)}{\|w\|}, \quad \text{s.t. } \forall i,\ y_i(w^\top x_i + b) \geq y_0(w^\top x_0 + b). \tag{3.20}$$

We claim that we can make $y_0(w^\top x_0 + b)$ arbitrarily large without affecting the result.

Remark

This is immediate: scaling w and b by the same factor λ does not change the optimization, since the numerator $y_i(w^\top x_i + b)$ and the denominator $\|w\|$ both scale by λ, and the factor cancels during simplification.

Thus, we may set $y_0(w^\top x_0 + b) = 1$. Define the *functional margin* as $y_i(w^\top x_i + b)$ and the *geometric margin* as $y_i(w^\top x_i + b)/\|w\|$. Since the functional margin alone has no intrinsic meaning, it is natural to rescale it by setting the norm of the support vector to unity.

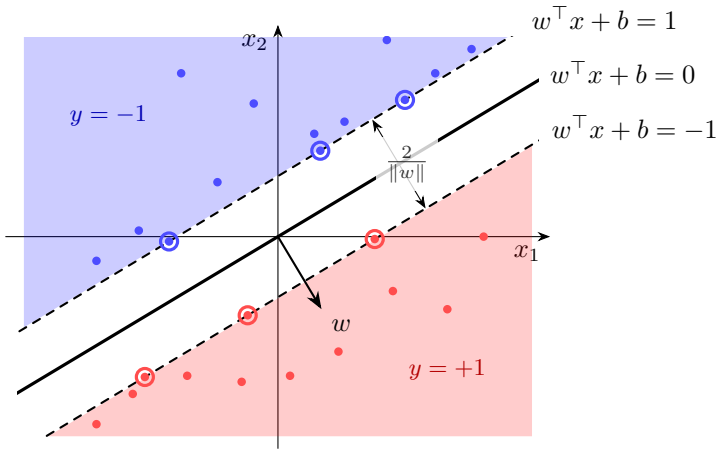


Figure 3.3: Support Vector Machine.

This yields the primal form of the SVM optimization problem:

$$\underset{w,b}{\operatorname{argmax}} \frac{1}{\|w\|}, \quad \text{s.t. } y_i(w^\top x_i + b) \geq 1 \quad \forall i \tag{3.21}$$

which is equivalent to

$$\underset{w,b}{\operatorname{argmin}} \frac{1}{2}\|w\|^2, \quad \text{s.t. } y_i(w^\top x_i + b) \geq 1 \quad \forall i \tag{3.22}$$

In other words, the goal is to separate the dataset correctly while ensuring a sufficient margin, and simultaneously minimizing the norm of w. This principle is known as **Structural Risk Minimization** (SRM).

Remark

Unlike logistic regression, datapoints far from separating hyperplane do not contribute loss. That is, only small subset of dataset is active as a constraint.

Proposition 3.1

SVM is a convex quadratic programming problem, which can be solved in polynomial time with standard packages. ♡

Problem 3.1

1. What if the dataset is not linearly separable?
2. What if γ too small because of outliers?

3.5 Dual Form Of SVM

3.5.1 Preliminaries from Convex Analysis*

⊛ ***This subsection is included out of personal interest and is not part of the course material. Readers who wish to focus solely on the course content may safely skip to the next subsection.***

Given a constrained optimization problem

$$\begin{aligned} \underset{x\in\mathbb{R}^n}{\text{argmin}} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \le 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned} \tag{3.23}$$

we define the *Lagrangian function* as

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \nu_j h_j(x), \tag{3.24}$$

where $\lambda_i \ge 0$ are the Lagrange multipliers associated with inequality constraints, and $\nu_j \in \mathbb{R}$ correspond to equality constraints.

Definition 3.2 (Lagrangian Dual Function)

The dual function *of a lagrangian function is defined by taking the infimum over the primal variable:*

$$\theta(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu). \tag{3.25}$$

♣

The function $\theta(\lambda, \nu)$ provides a lower bound on the optimal value of the primal problem, because for any feasible x and $\lambda \ge 0$,

$$\theta(\lambda, \nu) \le f(x). \tag{3.26}$$

Definition 3.3 (Lagrangian Dual Problem)

The dual problem *seeks the tightest possible lower bound on the primal optimum by maximizing $\theta(\lambda, \nu)$:*

$$\begin{aligned} \underset{\lambda,\nu}{\text{argmax}} \quad & \theta(\lambda, \nu) \\ \text{s.t.} \quad & \lambda_i \ge 0, \quad i = 1, \dots, m. \end{aligned} \tag{3.27}$$

Equivalently,

$$\underset{\lambda\ge 0,\nu}{\text{argmax}} \ \inf_x \ \mathcal{L}(x, \lambda, \nu). \tag{3.28}$$

♣

This formulation swaps the order of minimization and maximization, producing a problem in terms of the dual variables (λ, ν) instead of x. The resulting problem is always a *concave maximization*, even if the primal is nonconvex.

Theorem 3.5 (Weak Duality)

For any feasible x and any $\lambda \ge 0$, $\nu \in \mathbb{R}^p$, we have

$$\theta(\lambda, \nu) \le f(x). \tag{3.29}$$

Consequently, if p^* and d^* denote the optimal values of the primal and dual problems respectively, then

$$d^* \leq p^*. \tag{3.30}$$

Proposition 3.2 (Slater's Condition)

Consider the convex optimization problem:

$$\begin{aligned} \underset{x \in \mathbb{R}^n}{\text{argmin}} \quad & f(x) \\ s.t. \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & h_j(x) = 0, \quad j = 1, \ldots, p, \end{aligned} \tag{3.31}$$

where f and all g_i are convex functions, and each h_j is affine.
If there exists a point $\bar{x}$ (called a strictly feasible point*) such that*

$$g_i(\bar{x}) < 0, \quad i = 1, \ldots, m, \qquad h_j(\bar{x}) = 0, \quad j = 1, \ldots, p, \tag{3.32}$$

*then the problem satisfies **Slater's condition**.*

Slater's condition ensures that the feasible region of a convex optimization problem is not degenerate. It requires that there exists at least one point that *strictly* satisfies all inequality constraints (that is, $x \in \text{Int}(C)$).

Geometrically, this means the feasible set has a *non-empty interior*. If such a point $\bar{x}$ exists, the constraint surfaces do not touch or overlap in a pathological way, and the optimizer has enough space to operate smoothly. Under this mild regularity assumption, the duality gap between the primal and dual problems vanishes, leading to **strong duality**.

Theorem 3.6 (Strong Duality)

If the primal problem is convex and satisfies the Slater condition *(i.e., there exists a strictly feasible point x such that $g_i(x) < 0$ and $h_j(x) = 0$), then*

$$d^* = p^*. \tag{3.33}$$

That is, the dual problem attains the same optimal value as the primal problem.

Proof

Consider the convex optimization problem:

$$\begin{aligned} \underset{x \in \mathbb{R}^n}{\text{argmin}} \quad & f(x) \\ s.t. \quad & Ax \leq b, \\ & Cx = d, \end{aligned} \tag{3.34}$$

where $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a closed and convex function, and the constraints are linear.
The Lagrangian is defined as

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top (Ax - b) + \nu^\top (Cx - d), \qquad \lambda \geq 0. \tag{3.35}$$

The dual function is given by

$$g(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu), \tag{3.36}$$

and the dual problem is

$$\underset{\lambda \geq 0, \nu}{\text{argmax}} \; g(\lambda, \nu). \tag{3.37}$$

Weak duality always holds:

$$g(\lambda, \nu) \leq f(x), \qquad \forall x \text{ feasible}, \; \lambda \geq 0, \nu. \tag{3.38}$$

Hence, $d^* \leq p^*$.

Now, suppose there exists a strictly feasible point $\bar{x}$ such that

$$A\bar{x} < b, \qquad C\bar{x} = d. \tag{3.39}$$

This satisfies the Slater's condition. By the strong duality theorem *in convex optimization (also derived from Fenchel–Rockafellar duality), there exists a saddle point (x^*, λ^*, ν^*) such that*

$$\mathcal{L}(x^*, \lambda, \nu) \le \mathcal{L}(x^*, \lambda^*, \nu^*) \le \mathcal{L}(x, \lambda^*, \nu^*), \qquad \forall x, \lambda \ge 0, \nu. \tag{3.40}$$

From this saddle-point condition, the minimax equality follows:

$$\operatorname*{argmin}_{x} \operatorname*{argmax}_{\lambda \ge 0, \nu} \mathcal{L}(x, \lambda, \nu) = \operatorname*{argmax}_{\lambda \ge 0, \nu} \operatorname*{argmin}_{x} \mathcal{L}(x, \lambda, \nu), \tag{3.41}$$

which implies

$$p^* = d^*. \tag{3.42}$$

Hence, strong duality holds. ■

Remark

Strong duality implies that solving the dual problem is equivalent to solving the primal one, often with additional computational and theoretical advantages, as in the case of Support Vector Machines (SVMs).

Lemma 3.1 (Gordan's Lemma)

Define the function

$$f(x) = \ln \sum_{i=1}^{m} e^{\langle a_i, x \rangle}, \tag{3.43}$$

where $\{a_i\}_{i=1}^m$ are vectors in $\mathbb{R}^n$. The following statements are equivalent:

(a) $f(x)$ is bounded from below;

(b) there exists $\lambda \in \mathbb{R}^m$, $\lambda \ge 0$, $\lambda \ne 0$, such that

$$\sum_{i=1}^{m} \lambda_i a_i = 0; \tag{3.44}$$

(c) there exists no x satisfying

$$\langle a_i, x \rangle < 0, \quad i = 1, \dots, m. \tag{3.45}$$

♡

Moreover, the result can be generalized to a broader class of problems that also involve equality constraints.

Theorem 3.7 (Motzkin's transposition theorem)

Let $\{a_i\}_{i=1}^l$, $\{b_j\}_{j=1}^m$, and $\{c_k\}_{k=1}^p$ be vectors in $\mathbb{R}^n$, and let $\{\alpha_i\}_{i=1}^l$, $\{\beta_j\}_{j=1}^m$, and $\{\gamma_k\}_{k=1}^p$ be scalars. Then the linear system

$$\begin{aligned} \langle a_i, x \rangle &< \alpha_i, \quad i = 1, \dots, l, \\ \langle b_j, x \rangle &\le \beta_j, \quad j = 1, \dots, m, \\ \langle c_k, x \rangle &= \gamma_k, \quad k = 1, \dots, p, \end{aligned} \tag{3.7}$$

is inconsistent if and only if there exist vectors (multipliers)

$$\lambda_0 \in \mathbb{R}, \quad \lambda \in \mathbb{R}^l, \quad \mu \in \mathbb{R}^m, \quad \delta \in \mathbb{R}^p, \tag{3.46}$$

satisfying

$$\begin{aligned}\sum_{i=1}^{l}\lambda_i a_i + \sum_{j=1}^{m}\mu_j b_j + \sum_{k=1}^{p}\delta_k c_k = 0,\\ \sum_{i=1}^{l}\lambda_i \alpha_i + \sum_{j=1}^{m}\mu_j \beta_j + \sum_{k=1}^{p}\delta_k \gamma_k + \lambda_0 = 0,\\ (\lambda_0, \lambda, \mu) \geq 0, \qquad (\lambda_0, \lambda) \neq 0.\end{aligned} \tag{3.8}$$

Definition 3.4 (Hilbert Space)

*A **Hilbert space** is a complete inner product space.*

3.5.2 Dual Theory in the SVM Problem

Under the intuition provided by the Lagrangian formulation, we can define the primal objective as

$$p^* = \underset{w,b}{\operatorname{argmin}}\ \underset{\alpha\geq 0}{\operatorname{argmax}}\, L(w, b, \alpha), \tag{3.47}$$

where the Lagrangian L is defined as

$$L(w, b, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i\in[n]} \alpha_i(1 - y_i(w^\top x_i + b)) = \frac{1}{2}\|w\|^2 + \sum_{i\in[n]} \alpha_i - \sum_{i\in[n]} \alpha_i y_i(w^\top x_i + b). \tag{3.48}$$

We claim that:

Proposition 3.3

p^* corresponds to the solution of the primal SVM problem.

To justify this, suppose the claim were false. Assume that some constraint is not satisfied, that is, there exists an index i violating the constraint such that

$$1 - y_i(w^\top x_i + b) > 0. \tag{3.49}$$

Then the inner maximization problem

$$\theta_p(w, b) := \underset{\alpha\geq 0}{\operatorname{argmax}}\, L(w, b, \alpha) \tag{3.50}$$

can be driven to infinity, since the corresponding Lagrange multiplier $\alpha_i > 0$ can be increased arbitrarily to make $L(w, b, \alpha)$ unbounded above. Consequently,

$$\underset{w,b}{\operatorname{argmin}}\ \theta_p(w, b) = \underset{w,b}{\operatorname{argmin}}\ \infty \tag{3.51}$$

has no valid solution. Therefore, the initial assumption must be false, and the optimal dual variables α^* enforce all constraints to hold with equality or non-violation. Hence, the dual solution indeed corresponds to the optimal solution of the primal problem.

Consider a convex region defined by the constraint set C. We claim that for all $x_i \in \operatorname{Int}(C)$, that is, points located strictly inside the feasible region where all constraints are satisfied, the corresponding α_i is always set to 0. For all $x_i \in \partial C = \bar{C} - \operatorname{Int}(C)$, where the constraints become active, α_i may take positive values.

As a conclusion, regardless of which region the data points fall into, the following condition holds at the optimal solution due to the complementary slackness property in the KKT conditions:

$$\sum_{i\in[n]} \alpha_i\big(1 - y_i(w^\top x_i + b)\big) = 0. \tag{3.52}$$

This does not represent an algebraic identity but a structural condition satisfied at optimality. It implies that only the points lying exactly on the margin (the support vectors) contribute to the decision boundary, while all others have $\alpha_i = 0$.

Therefore, at the optimum, the objective effectively reduces to minimizing $\|w\|^2$.

Define the dual problem of p^* as follows:

$$d^* = \underset{\alpha \geq 0}{\operatorname{argmax}} \ \underset{w,b}{\operatorname{argmin}} L(w, b, \alpha), \tag{3.53}$$

where p^* is a min–max problem, while the dual problem is formulated as a max–min problem.

Theorem 3.8 (Weak Duality)

$$d^* \leq p^*. \tag{3.54}$$

Proof

By the definition of the argmin *(or equivalently, the infimum), we have*

$$d^* = \underset{\alpha \geq 0}{\operatorname{argmax}} \ \underset{w,b}{\operatorname{argmin}} L(w, b, \alpha) \ \leq \ \underset{\alpha \geq 0}{\operatorname{argmax}} \ L(w, b, \alpha) \tag{3.55}$$

for every (w, b).

We may then choose the best (w, b) *that minimizes the right-hand side, i.e.,*

$$d^* \leq \underset{w,b}{\operatorname{argmin}} \ \underset{\alpha \geq 0}{\operatorname{argmax}} L(w, b, \alpha) = p^*. \tag{3.56}$$

Weak duality always holds because no additional assumptions or constraints have been introduced in the argument above.

Here is another proof of the weak duality theorem:

Proof

Assume that (w_p, b_p) *is the optimal solution to the primal problem, and that* α_d *is the optimal solution to the dual problem. Then we have*

$$\begin{aligned} p^* &= \underset{w,b}{\operatorname{argmin}} \ \underset{\alpha \geq 0}{\operatorname{argmax}} L(w, b, \alpha) = \underset{\alpha \geq 0}{\operatorname{argmax}} \ L(w_p, b_p, \alpha) \\ &\geq L(w_p, b_p, \alpha_d) \geq \underset{w,b}{\operatorname{argmin}} \ L(w, b, \alpha_d) \\ &= \underset{\alpha \geq 0}{\operatorname{argmax}} \ \underset{w,b}{\operatorname{argmin}} \ L(w, b, \alpha) = d^*. \end{aligned}$$

Remark

We study the dual problem because, in many cases, it is analytically or computationally easier to solve than the original primal problem.

Weak duality only constrains the lower bound of the original problem. Is there a better situation for d^* if we include more conditions? The answer is yes. Click here for more information.

We claim that, for convex objectives, if all the constraints are linear functions, then the **Strong Duality** holds, where

$$d^* = p^*. \tag{3.57}$$

Remark

For completeness, I provide a brief proof in the previous subsection.

In the strong duality case, the inequalities in the second proof of the weak duality theorem become equalities. That is,

the intermediate solutions (w, b) in the dual problem are also optimal for the primal problem. Equivalently, the solutions of the dual and primal problems are **consistent** under strong duality.

Now, lets explicitly solve the dual problem under strong duality.

Solution

$$\operatorname*{argmax}_{\alpha\geq 0} \operatorname*{argmin}_{w,b} L(w,b,\alpha) = \operatorname*{argmax}_{\alpha\geq 0} \operatorname*{argmin}_{w,b} \left(\frac{1}{2}\|w\|^2 + \sum_{i\in[n]} \alpha_i(1 - y_i(w^\top x_i + b))\right) \tag{3.58}$$

To find the inner minimization over w, b, we apply the KKT stationarity conditions:

$$\frac{\partial L}{\partial w} = w - \sum_{i\in[n]} \alpha_i y_i x_i = 0 \Rightarrow \quad w = \sum_{i\in[n]} \alpha_i y_i x_i, \tag{3.59}$$

$$\frac{\partial L}{\partial b} = -\sum_{i\in[n]} \alpha_i y_i = 0 \Rightarrow \quad \sum_{i\in[n]} \alpha_i y_i = 0. \tag{3.60}$$

Substitute back into $L(w, b, \alpha)$. Plugging $w = \sum_i \alpha_i y_i x_i$ into L, we obtain:

$$L(w,b,\alpha) = \frac{1}{2}\|w\|^2 + \sum_{i\in[n]} \alpha_i(1 - y_i(w^\top x_i + b)) \tag{3.61}$$

$$= \frac{1}{2}\left\|\sum_{i\in[n]} \alpha_i y_i x_i\right\|^2 + \sum_{i\in[n]} \alpha_i - \sum_{i\in[n]} \alpha_i y_i \left(\sum_{j\in[n]} \alpha_j y_j x_j^\top x_i + b\right). \tag{3.62}$$

which follows the constraints:

$$\alpha \geq 0, \quad \sum_i \alpha_i y_i = 0 \tag{3.63}$$

Simplify L using $\sum_i \alpha_i y_i = 0$, we have

$$L(w,b,\alpha) = \sum_{i\in[n]} \alpha_i - \frac{1}{2}\sum_{i\in[n]}\sum_{j\in[n]} \alpha_i\alpha_j y_i y_j x_i^\top x_j. \tag{3.64}$$

Thus, the SVM dual problem becomes:

$$\boxed{\begin{aligned} \operatorname*{argmax}_{\alpha\geq 0} \quad & W(\alpha) = \sum_{i\in[n]} \alpha_i - \frac{1}{2}\sum_{i,j\in[n]} \alpha_i\alpha_j y_i y_j x_i^\top x_j, \\ s.t. \quad & \sum_{i\in[n]} \alpha_i y_i = 0. \end{aligned}} \tag{3.65}$$

Let α^* be the optimization solution of the uppon problem. Under strong duality, the optimal α^* gives rise to the optimal w^* and b^* through:

$$w^* = \sum_{i\in[n]} \alpha_i^* y_i x_i, \qquad b^* = y_i - w^{*\top} x_i \quad \text{for any support vector } \alpha_i^* > 0. \tag{3.66}$$

We can confirm whether (x_i, y_i) is support vector by checking whether α_i^* is positive (o.w. is zero). Thus:

$$w^* = \sum_{i\in \text{S.V.}s} \alpha_i^* y_i x_i, \qquad b^* = y_i - w^{*\top} x_i \quad \text{for any support vector } \alpha_i^* > 0. \tag{3.67}$$

Note

At this point, the reasoning effectively reproduces Gordan's Lemma within the derivation process. I provided a brief discussion in the previous subsection.

Thus, an SVM model can be fully reconstructed from its support vectors and their corresponding coefficients α_i^*. For any test point x, the prediction depends solely on the support vectors and their weights:

$$f(x) = w^{*\top} x + b^* = \sum_{i\in \text{S.V.}} \alpha_i^* y_i \, (x_i^\top x) + b^* > 0. \tag{3.68}$$

The red term in the equation above measures the *similarity* between the test point x and each support vector x_i.

To capture non-linear relationships, we can replace this inner product with a more general **kernel function** $K(x_i, x)$, thereby mapping the data into a higher-dimensional feature space while preserving computational efficiency. We will discuss kernel tricks in the next section.

We now introduce the practical algorithm commonly used in training SVMs: the **Sequential Minimal Optimization** (SMO) method. The main idea is to iteratively update a small subset of dual variables α_i while keeping all others fixed.

However, because of the equality constraint

$$\sum_{i=1}^{n} \alpha_i y_i = 0, \tag{3.69}$$

if we fix $(n-1)$ of the α_j, the remaining one α_i is automatically determined, leaving no degree of freedom. To overcome this, SMO randomly selects a pair (α_i, α_j) and optimizes them jointly while keeping all other α_k (for $k \neq i, j$) fixed. Under this setup, the equality constraint becomes

$$\alpha_i y_i + \alpha_j y_j = -\sum_{k \neq i,j} \alpha_k y_k = \text{constant}. \tag{3.70}$$

Hence, α_j can be expressed as a linear function of α_i, and the original optimization problem reduces to a one-dimensional quadratic programming problem in α_i, which is convex (positive semidefinite).

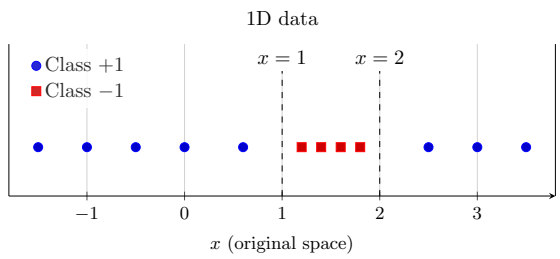
This reduction admits a **closed-form solution**, allowing efficient updates of the chosen pair (α_i, α_j).

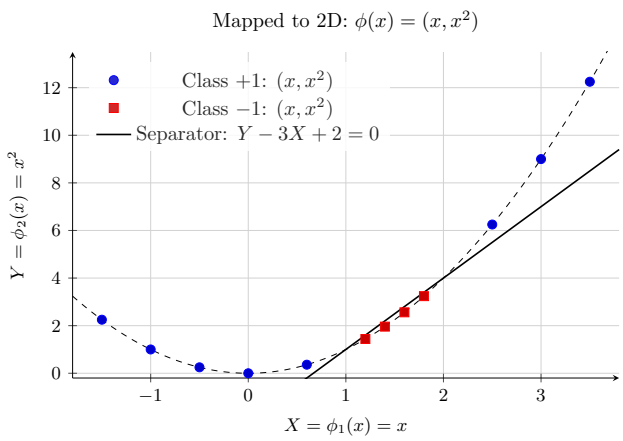
By repeating this process iteratively over different pairs, the algorithm converges to the optimal solution of the dual problem.

3.6 Kernel

In the previous section, we only considered linearly separable datasets, which is not always satisfied in general cases.

For example, consider a one-dimensional separation problem that cannot be separated by a single point (a one-dimensional plane); we can introduce an extra dimension to transform the dataset into a linearly separable one.





That is, we want to find a function φ that maps $\mathbb{R}^d \mapsto \mathbb{R}^{d'}$ to help the separation machine complete its work. For instance, for a two-dimensional input (x_1, x_2), we can define the mapping

$$\varphi(x) := (1, x_1, x_2, x_1^2, x_2^2, x_1 x_2) \in \mathbb{R}^6, \tag{3.71}$$

which makes several implicit features in $\mathbb{R}^2$ explicit.

We can then calculate the similarity between x_1 and x_2 through their transformed features $\varphi(x)$ in (3.68):

$$f(x) = w^{*\top} x + b^* = \sum_{i \in \text{S.V.}} \alpha_i^* y_i (\varphi(x_i)^\top \varphi(x)) + b^* > 0. \tag{3.72}$$

Correspondingly, for (3.64), the dual objective becomes

$$L = \sum_{i\in[n]} \alpha_i - \frac{1}{2} \sum_{i,j\in[n]} \alpha_i \alpha_j y_i y_j \langle \varphi(x_i), \varphi(x_j) \rangle. \tag{3.73}$$

Calculating the inner product in the high-dimensional feature space may be computationally expensive, as it often requires complex and high-order operations. Therefore, we define a **kernel function** that allows us to compute the inner product in the high-dimensional space indirectly, using only the original input vectors:

$$K(x_i, x_j) := \langle \varphi(x_i), \varphi(x_j) \rangle. \tag{3.74}$$

Definition 3.5 (Kernel Function)

*A function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a **kernel function** if there exists a Hilbert space $\mathcal{H}$ and a mapping*

$$\phi : \mathcal{X} \to \mathcal{H} \tag{3.75}$$

such that for all $x, x' \in \mathcal{X}$,

$$K(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{H}}. \tag{3.76}$$

♣

In the previous example, the kernel corresponding to $\varphi(x)$ can be written as

$$K(x, z) = (x^\top z + 1)^2. \tag{3.77}$$

We can easily verify this claim:

$$\begin{aligned} K(x, z) &= (x_1 z_1 + x_2 z_2 + 1)^2 \\ &= \langle (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1 x_2),\ (1, \sqrt{2}z_1, \sqrt{2}z_2, z_1^2, z_2^2, \sqrt{2}z_1 z_2) \rangle. \end{aligned}$$

Thus, we have effectively reduced the computational complexity from $O(d')$ to $O(d)$, thereby saving computational resources.

The dual formulation in (3.73) can then be simplified using the kernel function:

$$L = \sum_{i\in[n]} \alpha_i - \frac{1}{2} \sum_{i,j\in[n]} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \tag{3.78}$$

We now consider a new question: what kind of function K can be regarded as a valid kernel? Mercer's Theorem provides the precise answer to this problem.

Theorem 3.9 (Mercer's Theorem (Finite case))

A symmetric function $K(x, x')$ is a valid kernel if and only if, for any finite set $\{x_1, \ldots, x_n\} \subset \mathcal{X}$, the Gram matrix

$$\mathbf{K}_{ij} = K(x_i, x_j) \tag{3.79}$$

is positive semidefinite, i.e.

$$\forall c \in \mathbb{R}^n, \quad c^\top \mathbf{K} c \geq 0. \tag{3.80}$$

♡

Proof

In the finite-dimensional case.Let $\{x_1, \ldots, x_n\} \subset \mathcal{X}$ be any finite subset, and define the Gram matrix $\mathbf{K} \in \mathbb{R}^{n\times n}$ by

$$\mathbf{K}_{ij} = K(x_i, x_j), \qquad i, j = 1, \ldots, n. \tag{3.81}$$

Since K is symmetric, we have $\mathbf{K} = \mathbf{K}^\top$.
By the spectral theorem for real symmetric matrices, there exists an orthonormal basis of eigenvectors $\{v_1, \ldots, v_n\}$ and corresponding real eigenvalues $\lambda_1, \ldots, \lambda_n$ such that

$$\mathbf{K} = V \Lambda V^\top, \tag{3.82}$$

where $V = [v_1, \ldots, v_n]$ is orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

The kernel K being positive semidefinite implies that for all $c \in \mathbb{R}^n$,

$$c^\top \mathbf{K} c = \sum_{i,j} c_i K(x_i, x_j) c_j \geq 0, \tag{3.83}$$

which in turn implies $\lambda_i \geq 0$ for all i.
Now, define feature functions

$$\phi_i(x_j) = \sqrt{\lambda_i}\, v_i(j), \tag{3.84}$$

i.e. the j-th component of v_i scaled by $\sqrt{\lambda_i}$. Then for any $x_p, x_q \in \{x_1, \ldots, x_n\}$,

$$K(x_p, x_q) = \mathbf{K}_{pq} = \sum_{i=1}^{n} \lambda_i v_i(p) v_i(q) = \sum_{i=1}^{n} \phi_i(x_p)\, \phi_i(x_q). \tag{3.85}$$

Hence, in the finite-dimensional case, K can be written as an inner product

$$K(x_p, x_q) = \langle \phi(x_p), \phi(x_q) \rangle, \qquad \phi(x) = (\phi_1(x), \ldots, \phi_n(x))^\top. \tag{3.86}$$

This directly establishes the Mercer expansion in the finite case, and the general (infinite-dimensional) theorem follows by taking the limit as $n \to \infty$. ■

Theorem 3.10 (Mercer's Theorem)

Let $\mathcal{X} \subset \mathbb{R}^n$ be a compact set, and let $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a continuous, symmetric function.
*Then K is a **kernel** if and only if there exists an orthonormal basis $\{\phi_i\}_{i=1}^{\infty}$ of $L^2(\mathcal{X})$ and a sequence of non-negative eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$, $\lambda_i \geq 0$, such that for all $x, x' \in \mathcal{X}$,*

$$K(x, x') = \sum_{i=1}^{\infty} \lambda_i\, \phi_i(x)\, \phi_i(x'), \tag{3.87}$$

where the series converges absolutely and uniformly on $\mathcal{X} \times \mathcal{X}$. That is, K is positive semi-definite.
Equivalently, for any square-integrable function $f \in L^2(\mathcal{X})$,

$$\iint_{\mathcal{X}\times\mathcal{X}} f(x) K(x, x') f(x')\, dx\, dx' = \sum_{i=1}^{\infty} \lambda_i \Big(\int_{\mathcal{X}} f(x) \phi_i(x)\, dx \Big)^2 \geq 0. \tag{3.88}$$

♡

Here are some commonly used kernels:

1. **Linear kernel**: simply applies the inner product in the original space,

$$K(x, z) = x^\top z; \tag{3.89}$$

2. **Polynomial kernel**: maps $\mathbb{R}^n \to \mathbb{R}^{\binom{n+p}{p}}$,

$$K(x, z) = (x^\top z + 1)^p; \tag{3.90}$$

3. **Gaussian / RBF kernel (Radial Basis Function)**:

$$K(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2} \right). \tag{3.91}$$

Remark

The RBF kernel can be intuitively understood as measuring the similarity between two vectors based on their Euclidean distance.

The RBF kernel makes any dataset linearly separable, as it effectively maps x and z into an **infinite-dimensional** Hilbert space before taking their inner product, thus providing a powerful tool for nonlinear classification.

This property follows directly from the Taylor expansion of the exponential function, noting that the polynomial basis is complete.

Remark

$$\textit{Taylor Expansion:} \quad f(x) = f(0) + f'(0)\,x + \frac{1}{2!}f''(0)\,x^2 + \frac{1}{3!}f^{(3)}(0)\,x^3 + \cdots \tag{3.92}$$

$$k(x,z) = \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right)\exp\left(-\frac{\|z\|^2}{2\sigma^2}\right)\exp\left(\frac{x^\top z}{\sigma^2}\right) \tag{3.93}$$

$$= \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right)\exp\left(-\frac{\|z\|^2}{2\sigma^2}\right)\left[1 + \frac{x^\top z}{\sigma^2} + \frac{1}{2!}\left(\frac{x^\top z}{\sigma^2}\right)^2 + \frac{1}{3!}\left(\frac{x^\top z}{\sigma^2}\right)^3 + \cdots\right] \tag{3.94}$$

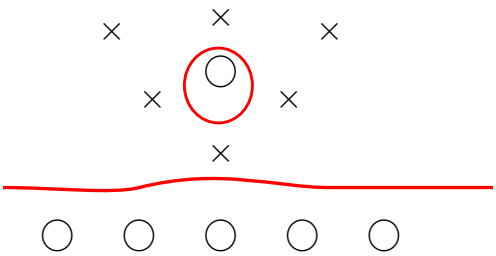
The only hyperparameter introduced here is σ^2, which determines the rate at which higher-order polynomial terms decay. A larger σ^2 implies a faster decay of these terms. Thus, σ^2 effectively controls the dimensionality of the feature space involved in the inner product.

Proposition 3.4

The RBF kernel enables nonlinear classification.

The above proposition follows from the behavior of the RBF kernel as $\sigma^2 \to 0^+$. In this limit, only the data point x_i closest to x contributes significantly to the kernel value, since its exponential decay is the slowest. In other words, the RBF kernel behaves almost like a **1-NN (nearest neighbor)** classifier in this regime.

To demonstrate the extraordinary capability of the RBF kernel, we note that it can represent **multiply connected**, or even **disconnected**, decision regions:



However, this expressive power also comes with a significant risk of **overfitting**.

We have now solved the non-linearly separable problem mentioned at the end of the last section, and we arrive at the next challenge: the presence of outliers.

To address this issue, we introduce the concept of **slack variables**. The idea is intuitive yet powerful — instead of forcing every data point to lie strictly on the correct side of the margin, we allow some samples to *violate* the margin constraint slightly, while penalizing these violations in the objective function.

Formally, we define non-negative variables

$$\xi_i \geq 0, \quad i = 1, \ldots, n, \tag{3.95}$$

called *slack variables*, representing the amount by which each sample fails to satisfy the margin condition. The optimization problem then becomes

$$\begin{aligned} \underset{w,b,\xi}{\operatorname{argmin}} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i, \\ \text{s.t.} \quad & y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad i = 1, \ldots, n, \\ & \xi_i \geq 0, \quad i = 1, \ldots, n. \end{aligned} \tag{3.96}$$

where $C > 0$ is a hyperparameter controlling the trade-off between maximizing the margin and minimizing classification

errors.

In this way, the model, known as the **Soft-Margin SVM**, allows small violations of the margin to achieve a better balance between robustness and generalization. This is the most commonly used form of SVM in practice.

We claim that the constraints on the slack variables are equivalent to the following formulation:

$$\xi_i \geq \max(0,\ 1 - y_i(w^\top x_i + b)). \tag{3.97}$$

In the minimization problem

$$\arg\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i, \tag{3.98}$$

it is crucial to observe that each ξ_i will always take the minimal feasible value,either 0 or $1 - y_i(w^\top x_i + b)$. Otherwise, if ξ_i were larger than necessary, the objective $\sum_i \xi_i$ could be further reduced by decreasing ξ_i.

Thus, the optimization problem can be reformulated as an unconstrained objective:

$$\operatorname*{argmin}_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \max(0,\ 1 - y_i(w^\top x_i + b)). \tag{3.99}$$

This formulation introduces the **Hinge loss**.

Definition 3.6 (Hinge loss)

The hinge loss function $L(z)$ is defined as:

$$L(z) = \max(0,\ 1 - z). \tag{3.100}$$

♣

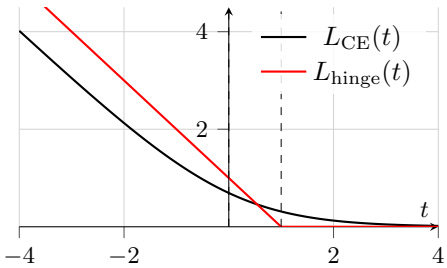


Figure 3.4: Hinge loss and CE loss.

Intuitively, the hinge loss $L_{\mathrm{hinge}}(t)$ remains zero when $t \geq 1$. This property directly enforces a margin: samples that already lie beyond the decision boundary (i.e., sufficiently far from the separating hyperplane) incur no loss, while those within or on the wrong side of the margin contribute a positive penalty. In this sense, the hinge loss explicitly encourages a wider margin between classes.

For comparison, consider the cross-entropy loss:

$$-\log(\sigma(z)) = \log(1 + e^{-z}), \tag{3.101}$$

where $\sigma(z)$ denotes the sigmoid function.

Remark

In class, we used $\log_2$ *instead of the natural logarithm, leading to an approximation when $x \to -\infty$:*

$$\log(1 + e^{-z}) \sim \frac{-z}{\log_2 e}, \tag{3.102}$$

which results in a slightly different scaling in the plotted curve.

Both the cross-entropy loss and the hinge loss serve as *substitute losses* for the zero-one loss:

$$L_{\text{zero-one}}(z) = \begin{cases} 1, & z < 0, \\ 0, & z \geq 0. \end{cases} \tag{3.103}$$

The zero-one loss is non-differentiable and provides no gradient information, so gradient-based optimization methods cannot be directly applied in this case.

Chapter 4 Representer Theorem

Introduction

- *Representer Theorem*
- *Dual form of Linear Regression*

4.1 Revisit the Preliminaries

We first provide a brief overview of the previous chapters on Support Vector Machines (SVM). The introduction of slack variables $\xi_i \geq 0,\ \forall i \in [n]$ allows the model to tolerate outliers or samples that do not strictly follow the shared distribution. However, it is important to properly scale the penalty on the slack variables, since inappropriate choices of this scaling factor C may severely affect the model's generalization ability.

The primal optimization problem of the soft-margin SVM can be written as:

$$\begin{aligned} &\operatorname*{argmin}_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \\ \text{s.t.} \quad & y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i \in [n]. \end{aligned}$$

We can further express the decision function of the SVM as:

$$f(x) = w^\top \varphi(x), \tag{4.1}$$

where $\varphi(x)$ denotes a feature mapping that transforms the input x into a (possibly higher-dimensional) feature space.

The objective function can then be rewritten as:

$$\operatorname*{argmin}_{w} \sum_{i\in[n]} \mathrm{Hinge}\big(f(x_i)\big) + \frac{\lambda}{2}\|w\|^2. \tag{4.2}$$

Remark

There is a minor difference between the above expression and the original SVM formulation: the bias term b can be absorbed into w by augmenting the input vector with an additional dimension (e.g., appending $x_0 = 1$).

By rewriting the $\mathrm{Hinge}\big(f(x_i)\big)$ term in terms of the slack variables ξ_i, we can derive the dual formulation of the soft-margin SVM:

$$\operatorname*{argmax}_{\alpha\geq 0,\beta\geq 0} \ \operatorname*{argmin}_{w,b} \left[\sum_{i\in[n]}\xi_i + \frac{\lambda}{2}w^\top w + \sum_{i\in[n]}\alpha_i\big(1 - y_i w^\top\varphi(x_i) - \xi_i\big) - \sum_{i\in[n]}\beta_i\xi_i\right]. \tag{4.3}$$

Here, α_i, β_i are both Lagrange multiplier. The gradient of the uppon targer brings:

$$\left.\begin{aligned} \partial_w L = 0 \\ \partial_{\xi_i} L = 0 \end{aligned}\right\} \quad \Rightarrow \quad \begin{cases} w = \dfrac{1}{\lambda}\displaystyle\sum_{i\in[n]}\alpha_i y_i \varphi(x_i) \\ \alpha_i + \beta_i = 1 \end{cases} \tag{4.4}$$

Bring back to (4.3):

$$\text{Dual}: \frac{1}{2}\lambda\left(\frac{1}{\lambda}\right)^2 \sum_{i,j\in[n]}\alpha_i\alpha_j y_i y_j \varphi^\top(x_i)\varphi(x_j) - \sum_{i\in[n]}\alpha_i y_i \varphi^\top(x_i)\frac{1}{\lambda}\sum_{j\in[n]}\alpha_j y_j \varphi(x_j) + \sum_{i\in[n]}\alpha_i \tag{4.5}$$

By simplification and kernel tricks:

$$\text{Dual}: \operatorname*{argmax}_{\alpha} -\frac{1}{2\lambda}\sum_{i,j\in[n]}\alpha_i\alpha_j y_i y_j k(x_i, x_j) + \sum_{i\in[n]}\alpha_i \quad s.t.\ 0 \leq \alpha_i \leq 1 \tag{4.6}$$

It can be observed that the only difference between the dual problems of the soft-margin and hard-margin SVMs lies in the constraint on α_i. In the soft-margin formulation, each α_i is required to satisfy $\alpha_i \leq 1$ (which is equivalent to introducing $\beta_i \geq 0$ and enforcing $\alpha_i + \beta_i = 1$). This constraint limits the influence of any single support vector, preventing the model

from overemphasizing outliers or noisy samples.

Ultimately, the prediction function $f(x)$ can be reformulated as a linear combination of kernel evaluations between the input point x and the training data points x_i:

$$f(x) = \sum_{i=1}^{n} \alpha_i \, k(x_i, x), \tag{4.7}$$

where $k(x_i, x) = \varphi(x_i)^\top \varphi(x)$ denotes the kernel function.

In the previous part, we introduced the notion of the *feature space*. We now provide a new example, the **Ridge Regression**, to demonstrate that this concept is widely used in the field of machine learning.

The objective function of Ridge Regression is:

$$\operatorname*{argmin}_{w} \frac{1}{2} \sum_{i \in [n]} \big(y_i - w^\top \varphi(x_i)\big)^2 + \frac{1}{2} \lambda \, w^\top w =: L(w) \tag{4.8}$$

Taking the derivative with respect to w gives:

$$\frac{\partial L}{\partial w} = - \sum_{i \in [n]} \big(y_i - w^\top \varphi(x_i)\big) \varphi(x_i) + \lambda w = 0 \tag{4.9}$$

Rearranging terms yields:

$$w = \frac{1}{\lambda} \sum_{i \in [n]} \big(y_i - w^\top \varphi(x_i)\big) \varphi(x_i) \tag{4.10}$$

We can rewrite this as:

$$w = \sum_{i \in [n]} \alpha_i \, \varphi(x_i) \tag{4.11}$$

where $\alpha_i = \frac{1}{\lambda}\big(y_i - w^\top \varphi(x_i)\big)$.

Substituting this representation into the prediction function gives:

$$f(x) = w^\top \varphi(x) = \sum_{i \in [n]} \alpha_i \, k(x_i, x) \tag{4.12}$$

where $k(x_i, x) = \varphi(x_i)^\top \varphi(x)$ is the kernel function.

The shared form also occurs in the logistic regression problem.

The objective function of regularized logistic regression is:

$$\min_{w} \sum_{i \in [n]} \log\big(1 + e^{-y_i w^\top \varphi(x_i)}\big) + \frac{1}{2} \lambda \, w^\top w \tag{4.13}$$

Taking the derivative with respect to w gives:

$$\frac{\partial L}{\partial w} = \sum_{i \in [n]} \frac{e^{-y_i w^\top \varphi(x_i)}}{1 + e^{-y_i w^\top \varphi(x_i)}} (-y_i) \varphi(x_i) + \lambda w = 0 \tag{4.14}$$

Rearranging the terms, we can express w as:

$$w = \sum_{i \in [n]} \alpha_i \, \varphi(x_i) \tag{4.15}$$

where

$$\alpha_i = \frac{1}{\lambda} \frac{y_i e^{-y_i w^\top \varphi(x_i)}}{1 + e^{-y_i w^\top \varphi(x_i)}}. \tag{4.16}$$

Substituting this representation into the prediction function gives:

$$f(x) = w^\top \varphi(x) = \sum_{i \in [n]} \alpha_i \, k(x_i, x), \tag{4.17}$$

The above observation suggests a deeper connection between *representation* and the *kernel trick*. It shows that all the regression models we have discussed so far can be interpreted as special cases of the same underlying idea.

4.2 Representer Theorem

We have already defined the concept of a *Hilbert space* in Definition 3.4. Building upon that foundation, we now introduce the notion of a **Reproducing Kernel Hilbert Space (RKHS)**, which extends the idea of inner product spaces to functional spaces, allowing the kernel trick to be formally grounded in Hilbert space theory.

Definition 4.1 (Reproducing Kernel Hilbert Space (RKHS))

Let $\mathcal{H}$ be a Hilbert space of functions $f : \mathcal{X} \to \mathbb{R}$ defined on a nonempty set $\mathcal{X}$. If there exists a function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ such that:

1. *For every $x \in \mathcal{X}$, the function $k(\cdot, x) \in \mathcal{H}$.*
2. *For every $f \in \mathcal{H}$ and every $x \in \mathcal{X}$, the* reproducing property *holds:*

$$f(x) = \langle f,\, k(\cdot, x)\rangle_{\mathcal{H}}, \tag{4.18}$$

*then $\mathcal{H}$ is called a **Reproducing Kernel Hilbert Space**, and k is called its **reproducing kernel**.* ♣

Remark

*In fact, the above definition is quite intuitive once we recall that a Hilbert space differs from a general vector space primarily in its **completeness** under an **inner product**. In other words, a Hilbert space is a complete inner product space, which extends the familiar geometric structure of finite-dimensional Euclidean spaces to infinite dimensions.*

Note

The completeness of a Reproducing Kernel Hilbert Space (RKHS) refers to the property that every Cauchy sequence in the space converges to a limit that still lies within the space. In geometric terms, this means the RKHS is "closed under limits" with respect to its inner-product-induced norm.
Moreover, by a fundamental result in Hilbert space theory, every separable Hilbert space admits a complete orthonormal basis. *Hence, an RKHS possesses (possibly infinite-dimensional) orthonormal basis functions $\{e_i\}_{i=1}^{\infty}$ such that any function $f \in \mathcal{H}$ can be expressed as*

$$f = \sum_{i=1}^{\infty} \langle f, e_i\rangle_{\mathcal{H}}\, e_i, \tag{4.19}$$

where the series converges in the Hilbert norm. In the context of kernel methods, Mercer's theorem ensures that the kernel function itself can be decomposed as

$$k(x, x') = \sum_{i=1}^{\infty} \lambda_i\, \phi_i(x)\phi_i(x'), \tag{4.20}$$

where $\{\sqrt{\lambda_i}\, \phi_i\}$ forms a complete orthonormal basis in the RKHS.

Theorem 4.1 (Mercer's Theorem)

Let $k(x, x')$ be a continuous, symmetric, and positive semi-definite kernel function defined on a compact domain $\mathcal{X} \subset \mathbb{R}^d$. Then there exists an orthonormal basis of eigenfunctions $\{\phi_i\}_{i=1}^{\infty}$ in $L^2(\mathcal{X})$ and a sequence of non-negative eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$ such that

$$k(x, x') = \sum_{i=1}^{\infty} \lambda_i\, \phi_i(x)\, \phi_i(x'), \tag{4.21}$$

where the series converges absolutely and uniformly. ♡

This result shows that any Mercer kernel defines a compact, self-adjoint, and positive operator $\mathcal{T}_k$ on $L^2(\mathcal{X})$:

$$(\mathcal{T}_k f)(x) = \int_{\mathcal{X}} k(x, x')\, f(x')\, dx'. \tag{4.22}$$

The eigenfunctions $\{\phi_i\}$ and eigenvalues $\{\lambda_i\}$ then form the spectral decomposition of $\mathcal{T}_k$.

Proposition 4.1

The RKHS $\mathcal{H}_k$ associated with k can be realized as the completion of the span of the kernel sections

$$\mathcal{H}_0 = \operatorname{span}\langle\, k(x, \cdot) \mid x \in \mathcal{X}\,\rangle. \tag{4.23}$$

Each $k(x, \cdot)$ can itself be expanded in the eigenbasis $\{\phi_i\}$ as

$$k(x, \cdot) = \sum_{i=1}^{\infty} \lambda_i\, \phi_i(x)\, \phi_i(\cdot). \tag{4.24}$$

♡

Consequently, every function in the RKHS can be approximated arbitrarily well by finite linear combinations of kernel sections $k(x_i, \cdot)$, since these functions span a dense subspace of $\mathcal{H}_k$. That is,

$$f = \sum_{i=1}^{\infty} \alpha_i\, k(x_i, \cdot), \tag{4.25}$$

where the series converges in the Hilbert norm.

Note

I have not listed the properties of the inner product here, as they should be familiar to anyone with basic mathematical knowledge, although Prof. Zhang presented them in detail during class for the sake of completeness.

Briefly, an RKHS is a space in which the value of a function at any point can be expressed as an inner product between the function itself and the corresponding kernel function.

Note

This property of the kernel function bears a striking resemblance to the idea of tensor contraction. *In tensor analysis, a multivariate tensor can be transformed into a new object by contracting over one or more indices, effectively "fixing" certain variables while leaving others free. Likewise, in the context of RKHS, a kernel $k(x, x')$ can be viewed as a bivariate function that, once one argument (say x_i) is fixed, induces a new function*

$$k(x_i, \cdot) : x' \mapsto k(x_i, x') \tag{4.26}$$

in the RKHS. These kernel sections $k(x_i, \cdot)$ thus play a role analogous to partially contracted tensors: each retains one free variable and collectively they span the functional space.

Example 4.1

Consider $f \in \mathcal{H}$, where $\mathcal{H}$ is a Hilbert sapce, $f(x) = w^\top x$, $x, w \in \mathbb{R}^d$, then $k(z, x) = z^\top x$ is the reproducing kernel of $\mathcal{H} = \operatorname{span}\langle k(x, \cdot),\, x \in \mathbb{R}^d\rangle$ under inner product $\langle k(w, \cdot), k(\cdot, x)\rangle_{\mathcal{H}} := k(w, x)$.

Proof

$$\langle k(w,\cdot), k(\cdot,x)\rangle_{\mathcal{H}} = k(w,x) = w^\top x = f(x) \tag{4.27}$$

■

Example 4.2

General form. $f(x) = w^\top \varphi(x)$ and $w \in \operatorname{span}\langle \varphi(x) : x \in \mathbb{R}^d\rangle$, $\varphi(x)$ can be expand under polynomial basis. Prove $k(z,x) := \varphi^\top(z)\varphi(x)$ is a reproduce kernel under inner product $\langle k(w,\cdot), k(\cdot,x)\rangle_{\mathcal{H}} := k(w,x)$.

Proof

Assume φ^{-1} exist:

$$\langle k(\varphi^{-1}(w),\cdot), k(\cdot,x)\rangle_{\mathcal{H}} = w^\top \varphi(x) = f(x) \tag{4.28}$$

Otherwise, expand f with kernel basis:

$$\sum_{i=1}^{\infty} \langle \alpha_i k(x_i,\cdot), k(\cdot,x)\rangle_{\mathcal{H}} = \underbrace{\sum_{i=1}^{\infty} \alpha_i \varphi^\top(x_i)}_{:= w^\top} \varphi(x) = w^\top \varphi(x) = f(x) \tag{4.29}$$

■

We can demonstrated that:

Proposition 4.2

The inner product of RKHS must have the term of:

$$\langle k(z,\cdot), k(\cdot,x)\rangle = k(z,x) \tag{4.30}$$

♡

Proof

$$\sum_{i=1}^{\infty} \langle \alpha_i k(x_i,\cdot), k(\cdot,x)\rangle_{\mathcal{H}} \underbrace{=}_{\text{by RKHS's properties}} f(x) = \underbrace{\sum_{i=1}^{\infty} \alpha_i k(x_i,x)}_{\textit{by } f(x)\textit{'s def}} \tag{4.31}$$

Seperate the uppon formula by α_i:

$$\langle k(z,\cdot), k(\cdot,x)\rangle = k(z,x) \tag{4.32}$$

■

In a Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$, the norm of a function $f \in \mathcal{H}$ is induced by the inner product:

$$\|f\|_{\mathcal{H}}^2 = \langle f, f\rangle_{\mathcal{H}}. \tag{4.33}$$

For a linear function of the form

$$f(x) = w^\top x, \tag{4.34}$$

we have

$$\langle f, f\rangle_{\mathcal{H}} = \langle k(w,\cdot),\ k(w,\cdot)\rangle_{\mathcal{H}} = k(w,w) = w^\top w. \tag{4.35}$$

Thus, the RKHS norm of f coincides with the Euclidean norm of its weight vector w.

For a feature-mapped function

$$f(x) = w^\top \varphi(x), \tag{4.36}$$

its inner product in the RKHS is given by

$$\langle f, f\rangle_{\mathcal{H}} = \langle k(\varphi^{-1}(w), \cdot),\ k(\varphi^{-1}(w), \cdot)\rangle_{\mathcal{H}} = k(\varphi^{-1}(w), \varphi^{-1}(w)). \tag{4.37}$$

Using the reproducing property of the kernel, we have

$$k(\varphi^{-1}(w), \varphi^{-1}(w)) = \langle \varphi(\varphi^{-1}(w)),\ \varphi(\varphi^{-1}(w))\rangle_{\mathcal{H}} = \varphi(\varphi^{-1}(w))^\top \varphi(\varphi^{-1}(w)) = w^\top w. \tag{4.38}$$

Now we came to the core of this chapter.

Theorem 4.2 (Representer Theorem)

Let $\mathcal{H}$ be a Reproducing Kernel Hilbert Space (RKHS) over a set $\mathcal{X}$ with reproducing kernel $k(\cdot,\cdot)$. Consider the regularized empirical risk minimization problem of the form:

$$\inf_{f\in\mathcal{H}} J(f) := \sum_{i\in[n]} L(f(x_i, y_i)) + R(\|f\|_{\mathcal{H}}), \tag{4.39}$$

where $L : \mathbb{R}^2 \to \mathbb{R}$ is an arbitrary loss function depending only on the function values at the sample points $\{x_i\}_{i=1}^n$, and R is a regularization function that $[0,\infty) \to \mathbb{R}$.
Then the minimizer $f^\star \in \mathcal{H}$ exists and admits the finite representation

$$f^\star = \sum_{i=1}^n \alpha_i\, k(x_i, \cdot), \tag{4.40}$$

for some coefficients $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$.

Proof

Any $f \in \mathcal{H}$ can be decomposed into two orthogonal components:

$$f = f_\parallel + f_\perp, \tag{4.41}$$

where $f_\parallel$ lies in the subspace $\mathcal{H}_0 := \operatorname{span}\{k(x_i,\cdot)\}_{i=1}^n$ and $f_\perp$ is orthogonal to $\mathcal{H}_0$.
We have

$$f_\perp(x_i) = \langle f_\perp,\ k(\cdot, x_i)\rangle_{\mathcal{H}} = 0, \quad \forall i. \tag{4.42}$$

Hence the loss term $L(f(x_1), \ldots, f(x_n))$ depends only on $f_\parallel$. Moreover, by the Pythagorean theorem,

$$R(\|f\|_{\mathcal{H}}) = R\left(\sqrt{\|\sum_{i\in[n]} \alpha_i k(x_i,\cdot)\|_{\mathcal{H}} + \|f_\perp\|_{\mathcal{H}}}\right) \geq R\left(|\sum_{i\in[n]} \alpha_i k(x_i,\cdot)|\right) \tag{4.43}$$

Replacing f with $f_\parallel$ thus never increases the objective in (4.39), implying that an optimal solution must lie in $\mathcal{H}_0$.
Therefore, the minimizer $f^\star$ can be expressed as a finite kernel expansion:

$$f^\star = \sum_{i=1}^n \alpha_i\, k(x_i, \cdot). \tag{4.44}$$

The Representer Theorem carries a profound geometric interpretation. Although the feature mapping

$$\varphi : \mathcal{X} \to \mathcal{H} \tag{4.45}$$

may embed the data into a potentially infinite-dimensional manifold within the RKHS, the optimization of f does not require direct manipulation in this high-dimensional space.

Specifically, the theorem states that the optimal function takes the form

$$f^\star = \sum_{i=1}^n \alpha_i\, k(x_i, \cdot), \tag{4.46}$$

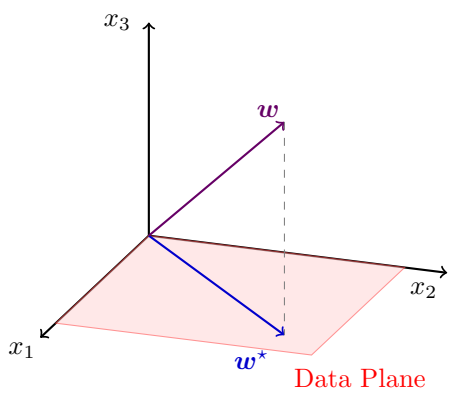
which lies entirely within the finite-dimensional submanifold

$$\mathcal{M}_n = \operatorname{span}\langle k(x_1, \cdot), k(x_2, \cdot), \ldots, k(x_n, \cdot)\rangle \subseteq \mathcal{H}. \tag{4.47}$$

Thus, even though $\mathcal{H}$ may be infinite-dimensional, the optimization problem effectively reduces to solving for the coefficients

$$\alpha_1,\ \alpha_2,\ \ldots,\ \alpha_n, \tag{4.48}$$

within this n-dimensional subspace.



Geometrically, the kernel trick enables us to perform optimization on a low-dimensional submanifold embedded in a high-dimensional feature manifold. Therefore, instead of optimizing a weight vector w in the full RKHS, we only need to search for the optimal combination of kernel coefficients $\{\alpha_i\}$ within the subspace spanned by the training data.

The above discussion reveals that a wide range of learning algorithms admit solutions expressed as weighted sums of kernel functions evaluated on finite training samples. This formulation transforms the prediction task into a problem of computing similarities between data points, rather than directly optimizing over high-dimensional parameters.

In this sense, kernel methods serve as a bridge between *parametric* and *non-parametric* models (such as k-Nearest Neighbors), combining the expressive flexibility of similarity-based approaches with the theoretical rigor of parameterized optimization frameworks.

4.3 Dual Form of Linear Regression

Consider the regularized least squares problem:

$$\min_{w}\ J(w) = \frac{1}{2}\sum_{i\in[n]}\left(y_i - w^\top\varphi(x_i)\right)^2 + \frac{1}{2}\lambda\, w^\top w, \tag{4.49}$$

where $\varphi(x_i)$ is the feature mapping and $\lambda > 0$ is a regularization coefficient. Taking the derivative of $J(w)$ with respect to w gives

$$\frac{\partial J}{\partial w} = \sum_{i\in[n]}\left(w^\top\varphi(x_i) - y_i\right)\varphi(x_i) + \lambda w = 0. \tag{4.50}$$

Rearranging terms, we can express w as a linear combination of feature vectors:

$$w = \sum_{i\in[n]}\alpha_i\,\varphi(x_i), \tag{4.51}$$

where $\alpha_i \in \mathbb{R}$ are the dual coefficients.

Let the feature matrix be

$$\Phi = \begin{pmatrix} \varphi(x_1)^\top \\ \varphi(x_2)^\top \\ \vdots \\ \varphi(x_n)^\top \end{pmatrix} \in \mathbb{R}^{n\times d}. \tag{4.52}$$

Then w can be compactly written as

$$w = \Phi^\top \alpha. \tag{4.53}$$

Substituting into the objective function yields:

$$J(w) = \frac{1}{2}(\Phi w - y)^\top (\Phi w - y) + \frac{1}{2}\lambda\, w^\top w. \tag{4.54}$$

By substituting $w = \Phi^\top \alpha$, we obtain:

$$J(\alpha) = \frac{1}{2}(\Phi\Phi^\top \alpha - y)^\top (\Phi\Phi^\top \alpha - y) + \frac{1}{2}\lambda\, \alpha^\top \Phi\Phi^\top \alpha. \tag{4.55}$$

Defining the kernel matrix (**Gram Matrix**):

$$K = \Phi\Phi^\top, \quad K_{ij} = \langle \varphi(x_i), \varphi(x_j) \rangle = k(x_i, x_j), \tag{4.56}$$

the dual objective becomes:

$$J(\alpha) = \frac{1}{2}(K\alpha - y)^\top (K\alpha - y) + \frac{1}{2}\lambda\, \alpha^\top K\alpha. \tag{4.57}$$

By derivation:

$$J(\alpha) = \frac{1}{2}\alpha^\top KK\alpha + \frac{1}{2}y^\top y - \alpha^\top Ky + \frac{\lambda}{2}\alpha^\top K\alpha \tag{4.58}$$

$$\frac{\partial J(\alpha)}{\partial \alpha} = KK\alpha - Ky + \lambda K\alpha = 0 \tag{4.59}$$

If $K \succ 0$:

$$(K + \lambda I)\alpha = y \quad \Rightarrow \quad \alpha = (K + \lambda I)^{-1} y \tag{4.60}$$

By simplification, notice that $w = \Phi^\top \alpha$, the prediction function can be expressed as:

$$f(x) = w^\top \varphi(x) = \begin{pmatrix} k(x_1, x) \\ \vdots \\ k(x_n, x) \end{pmatrix} (K + \lambda I)^{-1} y. \tag{4.61}$$

Chapter 5 Learning Theory

Introduction

- *In-Sample Error*
- *Out-of-Sample Error*
- *Generalization Error*
- *PAC*
- *Shelah's Lemma*
- *VC-Dim*

5.1 Learning Theory

Learning theory forms the foundation of machine learning. It explains why a model can generalize to data that did not appear during training. We may draw an analogy here:

- Training a model $\longrightarrow$ Taking a course;
- Training process $\longrightarrow$ Doing exercises or homework;
- Testing phase $\longrightarrow$ Taking an exam.

From our own learning experience, we know that students who perform well during practice do not always achieve high scores in exams. Likewise, a model that fits the training data well may still perform poorly on unseen data. This motivates us to introduce theoretical concepts that can quantitatively describe and explain such phenomena.

Definition 5.1 (In-sample Error)

The in-sample error E_{in} *is defined as the empirical error rate of a hypothesis evaluated on the training dataset.* ♣

Let $h \in \mathcal{H}$ denote a hypothesis or model, for example a linear classifier $h(x) := \text{sign}(w^\top x + b)$. Then the in-sample error can be formally written as

$$E_{\text{in}} = \frac{1}{n} \sum_{i \in [n]} \mathbb{1}\big(h(x_i) \neq y_i\big), \qquad D := \{(x_i, y_i) \mid i \in [n]\}. \tag{5.1}$$

Here D denotes the training dataset, and each $y_i \in \{-1, 1\}$ is the ground-truth label prepared for the classification task.

Oppositely, we also wish to evaluate the model's ability to perform well on unseen or more general tasks. This motivates the definition of a new concept:

Definition 5.2 (Out-of-sample Error)

The out-of-sample error E_{out} *measures how well a model generalizes beyond the training data.* ♣

In the previous example, E_{out} can be expressed as

$$E_{\text{out}} = \mathbb{P}\big(h(x) \neq y\big) = \mathbb{E}_{(x,y)\sim p_{xy}}\left[\mathbb{1}\big(h(x) \neq y\big)\right], \tag{5.2}$$

where $(x, y) \sim p_{xy}$ are i.i.d. samples drawn from the true underlying data distribution.

To quantitatively capture the generalization ability of a model, we define the following notion:

Definition 5.3 (Generalization Error)

The generalization error *is defined as the difference between the out-of-sample and in-sample errors:*

$$E_{\text{out}}(h) - E_{\text{in}}(h). \tag{5.3}$$

♣

Remark

In practice, we often introduce regularization terms to prevent E_{in} from becoming excessively small, which can help improve generalization performance.

Our ultimate goal is to bound the quantity $E_{\text{out}}(h) - E_{\text{in}}(h)$. However, it remains an open question whether such a bound can be feasibly obtained for arbitrary hypotheses.

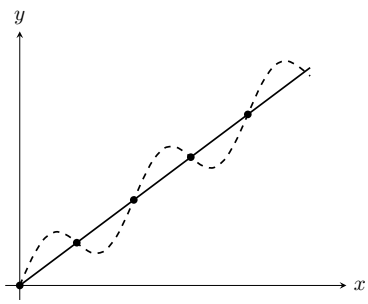


Figure 5.1: An example where $E_{\text{in}} = 0$ but E_{out} remains large.

It is crucial to acknowledge that, in essence, no finite dataset can guarantee a definitive prediction of future outcomes. The optimization of a model solely based on minimizing $E_{\text{in}}(h)$ does not ensure a small generalization gap $E_{\text{out}}(h) - E_{\text{in}}(h)$, which may in fact be arbitrarily large. **However**, according to probabilistic learning theory, we can establish that with high probability $(1 - \delta)$, where $\delta \ll 1$, the generalization error is bounded by a small quantity $\varepsilon(\mathcal{H}, n, \delta)$ depending on the hypothesis space $\mathcal{H}$, the number of samples n, and the confidence parameter δ:

$$\mathbb{P}(|E_{\text{out}}(h) - E_{\text{in}}(h)| \leq \varepsilon(\mathcal{H}, n, \delta)) \geq 1 - \delta. \tag{5.4}$$

This forms the cornerstone of statistical learning theory: while perfect certainty is unattainable, we can quantify uncertainty and guarantee performance with high probability.

The theorem above leads to the framework known as the **Probably Approximately Correct (PAC) Learning Theory**. Before introducing the PAC formulation in detail, we first review several fundamental mathematical tools.

Theorem 5.1 (Hoeffding's Inequality)

Let $X_1, \ldots, X_n$ be independent random variables such that each X_i is almost surely bounded, $X_i \in [a_i, b_i]$. Define the empirical mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then, for any $\varepsilon > 0$, we have

$$\mathbb{P}\big(|\bar{X} - \mathbb{E}\bar{X}| \geq \varepsilon\big) \leq 2\exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right). \tag{5.5}$$

♡

Proof

Let $S = \sum_{i=1}^{n} X_i$. We first bound the upper tail $\mathbb{P}(S - \mathbb{E}S \geq t)$; the lower tail is analogous and the two-sided bound follows by union.

By Markov's inequality, for any $\lambda > 0$,

$$\mathbb{P}(S - \mathbb{E}S \geq t) = \mathbb{P}\left(e^{\lambda(S-\mathbb{E}S)} \geq e^{\lambda t}\right) \leq e^{-\lambda t}\,\mathbb{E}\,e^{\lambda(S-\mathbb{E}S)}. \tag{5.6}$$

Because $X_1, \ldots, X_n$ are independent,

$$\mathbb{E}\,e^{\lambda(S-\mathbb{E}S)} = \prod_{i=1}^{n}\mathbb{E}\,e^{\lambda(X_i-\mathbb{E}X_i)}. \tag{5.7}$$

For each bounded random variable $Y \in [a, b]$ with $\mathbb{E}Y = \mu$, Hoeffding's lemma gives

$$\mathbb{E}\,e^{\lambda(Y-\mu)} \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \tag{5.8}$$

Applying this to each X_i yields

$$\mathbb{E}\, e^{\lambda(S-\mathbb{E}S)} \le \exp\left(\frac{\lambda^2}{8}\sum_{i=1}^{n}(b_i-a_i)^2\right). \tag{5.9}$$

Hence,

$$\mathbb{P}(S-\mathbb{E}S \ge t) \le \exp\left(-\lambda t + \frac{\lambda^2}{8}\sum_{i=1}^{n}(b_i-a_i)^2\right). \tag{5.10}$$

Optimizing over $\lambda > 0$, we find

$$\lambda^\star = \frac{4t}{\sum_{i=1}^{n}(b_i-a_i)^2}, \tag{5.11}$$

which gives

$$\mathbb{P}(S-\mathbb{E}S \ge t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i-a_i)^2}\right). \tag{5.12}$$

By symmetry,

$$\mathbb{P}(S-\mathbb{E}S \le -t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i-a_i)^2}\right). \tag{5.13}$$

Combining both tails gives

$$\mathbb{P}(|S-\mathbb{E}S| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i-a_i)^2}\right). \tag{5.14}$$

Finally, setting $t = n\varepsilon$ and noting $\bar{X} = S/n$, we obtain

$$\mathbb{P}\big(|\bar{X}-\mathbb{E}\bar{X}| \ge \varepsilon\big) \le 2\exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^{n}(b_i-a_i)^2}\right). \tag{5.15}$$

■

Now, for a **given and fixed** hypothesis h, we obtain the following result:

Theorem 5.2 (First Bound of PAC Learning)

For any $\varepsilon > 0$,

$$\mathbb{P}(E_{\text{out}}(h) - E_{\text{in}}(h) \ge \varepsilon) \le \exp(-2n\varepsilon^2). \tag{5.16}$$

♡

Here, $E_{\text{in}}(h)$ serves as the empirical mean of the random variables $\mathbb{1}\big(h(x_i) \neq y_i\big)$, as defined previously. In other words, each indicator $\mathbb{1}\big(h(x_i) \neq y_i\big)$ represents a bounded random variable in $[0,1]$, and Hoeffding's inequality ensures that the deviation between the empirical error $E_{\text{in}}(h)$ and the expected error $E_{\text{out}}(h)$ decreases exponentially with the number of samples n.

Proof

By taking expectation over the definition of E_{in}, we have

$$\begin{aligned}
\mathbb{E}[\bar{X}] &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_{(x_i,y_i)\sim p_{xy}}\big[\mathbb{1}(h(x_i) \neq y_i)\big] \\
&= \frac{1}{n}\sum_{i=1}^{n}E_{\text{out}}(h) = E_{\text{out}}(h).
\end{aligned}$$

The third equality follows directly from the definition of $E_{\text{out}}(h)$. Applying Hoeffding's inequality, note that each random variable $\mathbb{1}(h(x_i) \neq y_i)$ is bounded in $\{0,1\}$, so $(b_i-a_i)^2 = 1$ for all i, and the denominator $\sum_{i=1}^{n}(b_i-a_i)^2 = n$ cancels accordingly. This completes the proof. ■

However, there is a fundamental limitation in the condition of this first PAC bound: the hypothesis h must be fixed *before* observing the training data. In practice, the learning algorithm selects h based on the dataset D, which introduces statistical dependence between h and D. (Parameters of model h is determined by the chosen training dataset D) This dependence violates the independence assumption required by Hoeffding's inequality, and therefore the bound no longer holds directly once h is data-dependent.

To overcome this limitation, we turn to analyzing the **union bound** over the hypothesis space $\mathcal{H}$, which removes the dependency on a single fixed hypothesis chosen after observing the training data. We first assume that $\mathcal{H}$ is a finite set, which actually is not always satisfied in practice (e.g., the space of linear models has infinite cardinality, $|\mathrm{GL}_n| = \infty$). That is, let $\mathcal{H} = \{h_1, \ldots, h_M\}$.

In the worst case, we have

$$\begin{aligned}\mathbb{P}(\exists\, h \in \mathcal{H} : E_{\text{out}}(h) - E_{\text{in}}(h) \geq \varepsilon) &= \mathbb{P}\left(\bigcup_{i=1}^{M} \{E_{\text{out}}(h_i) - E_{\text{in}}(h_i) \geq \varepsilon\}\right) \\ &\underbrace{\leq}_{\text{Union Bound}} \sum_{i=1}^{M} \mathbb{P}(E_{\text{out}}(h_i) - E_{\text{in}}(h_i) \geq \varepsilon) \\ &\leq M \cdot \exp(-2n\varepsilon^2).\end{aligned}$$

Hence, we obtain the following result:

Theorem 5.3 (First Practical Bound of PAC Learning)

With probability at least $1 - \delta$,

$$E_{\text{out}}(h_i) - E_{\text{in}}(h_i) < \sqrt{\frac{1}{2n} \log \frac{M}{\delta}}. \tag{5.17}$$

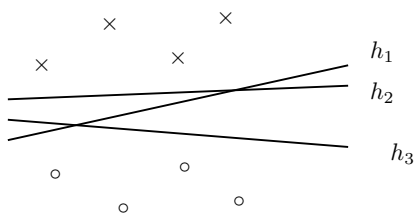
♡

Remark

$$\delta := M \cdot \exp(-2n\varepsilon^2) \quad \Leftrightarrow \quad \varepsilon = \sqrt{\frac{1}{2n} \log \frac{M}{\delta}} \tag{5.18}$$

This result reveals a fundamental trade-off between model complexity and generalization. As the hypothesis space $\mathcal{H}$ becomes richer, the total number of possible hypotheses M increases. A larger M expands the logarithmic term $\log \frac{M}{\delta}$, thereby loosening the bound on $E_{\text{out}} - E_{\text{in}}$. In other words, even though a highly expressive model may achieve $E_{\text{in}} = 0$ by perfectly fitting the training data, its generalization gap can still remain large. To maintain reliable generalization for such expressive models, one must increase the sample size n accordingly. This highlights the essential balance between data volume and model capacity in learning theory.

Now consider an infinite hypothesis space $\mathcal{H}$, for example the reproducing kernel Hilbert space (RKHS) of linear classifiers $\{h : h(x) = w^\top x + b\}$. We claim that the bound provided in the theorem above becomes an extreme overestimation in such cases. Intuitively, the union bound treats all hypotheses as completely distinct, even if many of them make identical predictions on the dataset. For example, the following three classifiers h_1, h_2, and h_3 may produce exactly the same classification results, yet the union bound still counts them as three separate events, thereby inflating the estimated probability by a factor of three.



Note

A natural idea is to quotient out an equivalence relation on $\mathcal{H}$ so that its effective measure decreases. This leads us to the concept of dichotomy*, where we seek an isomorphism that maps hypotheses producing identical classification results to the same image, thereby identifying them within a common kernel space. By quotienting out the equivalence classes induced by* ker*, which acts as a normal subgroup, we obtain a new structure—still a group (or more appropriately here, a reduced hypothesis space)— a space of smaller measure, serving as a kind of* renormalized *hypothesis space.*

Assume a binary classification setting

$$y \in \{-1, +1\}, \qquad x \in \mathcal{X}, \tag{5.19}$$

with hypothesis space

$$\mathcal{H} = \{h : \mathcal{X} \to \{-1, +1\}\}. \tag{5.20}$$

Given n samples

$$x_1, x_2, \ldots, x_n \in \mathcal{X}, \tag{5.21}$$

for any $h \in \mathcal{H}$ we obtain an n–tuple of labels

$$\big(h(x_1), h(x_2), \ldots, h(x_n)\big) \in \{-1, +1\}^n, \tag{5.22}$$

which is called a *dichotomy* of the set $\{x_1, \ldots, x_n\}$ induced by $\mathcal{H}$.

The growth function $m_{\mathcal{H}}(n)$ is defined as:

Definition 5.4 (Growth Function)

$$m_{\mathcal{H}}(n) = \max_{x_1,\ldots,x_n \in \mathcal{X}} \Big|\big\{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\big\}\Big|, \tag{5.23}$$

♣

i.e., the maximum number of distinct dichotomies that $\mathcal{H}$ can implement on any set of n points. It measures the effective size of the hypothesis space $\mathcal{H}$.

Note

The growth function can be interpreted as a quotient construction. Define the evaluation map

$$\Phi_n : \mathcal{H} \to \{-1, +1\}^n, \qquad \Phi_n(h) = (h(x_1), h(x_2), \ldots, h(x_n)). \tag{5.24}$$

This map induces an equivalence relation on $\mathcal{H}$:

$$h_1 \sim h_2 \quad \Leftrightarrow \quad h_1(x_i) = h_2(x_i),\ \forall i = 1, \ldots, n. \tag{5.25}$$

Each equivalence class corresponds to one unique dichotomy, so two hypotheses are considered identical if they generate the same labeling on the given sample set. Therefore, the growth function can be written as

$$m_{\mathcal{H}}(n) = \big|\mathcal{H}/\ker(\Phi_n)\big|, \tag{5.26}$$

representing the number of equivalence classes of $\mathcal{H}$ under this relation. Intuitively, this quotient construction mods out by the kernel of the evaluation map, focusing only on the empirically distinguishable behaviors of hypotheses on n samples.

Since there are at most 2^n different ± 1 labelings on n points, we always have

$$m_{\mathcal{H}}(n) \le 2^n. \tag{5.27}$$

Now let $S = \{x_1, \ldots, x_n\} \subset \mathcal{X}$ be a finite subset. We say that $\mathcal{H}$ *shatters* S if it can realize all possible ± 1 assignments on S, that is,

$$\mathcal{H}(x_1, \ldots, x_n) = \{-1, +1\}^n. \tag{5.28}$$

Equivalently, for every labeling $(y_1, \ldots, y_n) \in \{-1, +1\}^n$ there exists some $h \in \mathcal{H}$ such that $h(x_i) = y_i$ for all $i = 1, \ldots, n$.

Example 5.1

Consider three sample points x_1, x_2, x_3 and a hypothesis set $\mathcal{H}$ that induces the following dichotomies:

$$\mathcal{H}(x_1, x_2, x_3) = \{(+1, -1, -1),\ (-1, +1, -1),\ (-1, +1, +1)\}. \tag{5.29}$$

Determine which subsets of $\{x_1, x_2, x_3\}$ can be *shattered* by $\mathcal{H}$.

Solution

- *For any single point $\{x_i\}$, both $+1$ and -1 appear in the projections of $\mathcal{H}(x_1, x_2, x_3)$, so every singleton subset is shattered.*
- *For pairs of points:*

  $\{x_1, x_2\} \Rightarrow \{(+1, -1),\ (-1, +1)\}$ *(missing $(+1, +1)$ and $(-1, -1)$) $\Rightarrow$ not shattered;*

  $\{x_1, x_3\} \Rightarrow \{(+1, -1),\ (-1, +1),\ (-1, -1)\}$ *(missing $(+1, +1)$) $\Rightarrow$ not shattered;*

  $\{x_2, x_3\} \Rightarrow \{(-1, -1),\ (+1, -1),\ (+1, +1)\}$ *(all $2^2 = 4$ combinations not present) $\Rightarrow$ not shattered.*
- *For the full set $\{x_1, x_2, x_3\}$, since only $3 < 2^3 = 8$ dichotomies are realized, it is clearly not shattered.*

Hence, $\mathcal{H}$ can only shatter the singleton subsets $\{x_1\}$, $\{x_2\}$, *and* $\{x_3\}$, *but no pair or triple of points.*

Example 5.2

Let $\mathcal{X} = \mathbb{R}^2$, and consider the hypothesis set of all linear classifiers

$$\mathcal{H} = \{\, h_{w,b}(x) = \operatorname{sign}(w^\top x + b) \mid w \in \mathbb{R}^2,\ b \in \mathbb{R} \,\}. \tag{5.30}$$

Determine the value of $m_{\mathcal{H}}(3)$, the maximum number of dichotomies that $\mathcal{H}$ can produce on any 3 points in $\mathbb{R}^2$.

Solution

A separating hyperplane in $\mathbb{R}^2$ is a line of the form $w^\top x + b = 0$. The sign of $w^\top x + b$ assigns each point to one of two half-spaces.

For any given labeling $(y_1, y_2, y_3) \in \{-1, +1\}^3$, we can always find some line that places the $+1$ points on one side and the -1 points on the other, as long as the three points are not collinear. This is because two points determine a line, and the third point's label can always be satisfied by shifting or rotating that line slightly.

Hence every possible assignment of ± 1 labels to the three points is realizable:

$$\mathcal{H}(x_1, x_2, x_3) = \{-1, +1\}^3. \tag{5.31}$$

Therefore, the growth function value is

$$m_{\mathcal{H}}(3) = 2^3 = 8. \tag{5.32}$$

Remark

More examples have been presented in class, but I'm not goint to note these in our lecture.

$m_{\mathcal{H}}(n)$ quantified the ability of a model.

5.2 VC-Dim

Definition 5.5 (Vapnik–Chervonenkis Dimension)

The VC dimension $\mathrm{dvc}(\mathcal{H})$ *is the largest integer n such that*

$$m_{\mathcal{H}}(n) = 2^n. \tag{5.33}$$

♣

Remark

Vapnik is the inventor of the Support Vector Machine (SVM) and one of the founders of statistical learning theory.

The VC dimension measures the *capacity* or *expressive power* of a hypothesis class. Formally, it is the largest integer n such that $\mathcal{H}$ can realize all 2^n possible dichotomies on some set of n points. Equivalently, it quantifies how complex the decision boundaries in $\mathcal{H}$ can be: a larger VC dimension means $\mathcal{H}$ can represent more distinct labelings of data, i.e. greater capacity.

It turns out that if $\mathrm{dvc}(\mathcal{H})$ is finite, then $m_{\mathcal{H}}(n)$ can be bounded in terms of $\mathrm{dvc}(\mathcal{H})$.

Theorem 5.4 ((Sauer–)Shelah's Lemma)

Let $\mathcal{H}$ be a class of binary-valued functions on a domain $\mathcal{X}$. Define its growth function as

$$m_{\mathcal{H}}(n) = \max_{\substack{S \subseteq \mathcal{X} \\ |S|=n}} \left| \{ (h(x))_{x \in S} : h \in \mathcal{H} \} \right|. \tag{5.34}$$

If $\mathrm{dvc}(\mathcal{H}) = d < \infty$, *then for all* $n \geq 0$,

$$m_{\mathcal{H}}(n) \leq \sum_{i=0}^{d} \binom{n}{i} \sim O(n^d) \tag{5.35}$$

♡

Proof

We proceed by induction on n. The base case is trivial and we proved that in the next proof for completeness.

Step 1: Divide the hypothesis class. *Fix $x_1, \ldots, x_n \in \mathcal{X}$, and define*

$$\mathcal{H}_1(x_1, \ldots, x_{n-1}) = \{(h(x_1), \ldots, h(x_{n-1})) \mid h \in \mathcal{H},\ h(x_n) = +1\}, \tag{5.36}$$

$$\mathcal{H}_2(x_1, \ldots, x_{n-1}) = \{(h(x_1), \ldots, h(x_{n-1})) \mid h \in \mathcal{H},\ h(x_n) = -1\}. \tag{5.37}$$

Then

$$\mathcal{H}(x_1, \ldots, x_n) = \{(\mathbf{v}, +1) \mid \mathbf{v} \in \mathcal{H}_1(x_1, \ldots, x_{n-1})\} \cup \{(\mathbf{v}, -1) \mid \mathbf{v} \in \mathcal{H}_2(x_1, \ldots, x_{n-1})\}. \tag{5.38}$$

Step 2: Counting distinct dichotomies. *If $\mathbf{v}$ belongs to both $\mathcal{H}_1$ and $\mathcal{H}_2$, then $\mathcal{H}$ realizes both labelings $(\mathbf{v}, +1)$ and $(\mathbf{v}, -1)$. Hence,*

$$|\mathcal{H}(x_1, \ldots, x_n)| \leq |\mathcal{H}_1(x_1, \ldots, x_{n-1})| + |\mathcal{H}_2(x_1, \ldots, x_{n-1})|. \tag{5.39}$$

Step 3: Bounding the VC dimension. *If* $\mathrm{dvc}(\mathcal{H}) = d$, *then* $\mathrm{dvc}(\mathcal{H}_1) \leq d$ *and* $\mathrm{dvc}(\mathcal{H}_2) \leq d - 1$. *Otherwise, if $\mathcal{H}_2$ could shatter a subset of size d, then $\mathcal{H}$ could shatter that subset together with x_n, contradicting* $\mathrm{dvc}(\mathcal{H}) = d$.

Step 4: Apply the induction hypothesis. *By the inductive assumption,*

$$|\mathcal{H}_1(x_1, \ldots, x_{n-1})| \leq \sum_{i=0}^{d} \binom{n-1}{i}, \qquad |\mathcal{H}_2(x_1, \ldots, x_{n-1})| \leq \sum_{i=0}^{d-1} \binom{n-1}{i}. \tag{5.40}$$

Therefore,

$$|\mathcal{H}(x_1, \ldots, x_n)| \leq \sum_{i=0}^{d} \binom{n-1}{i} + \sum_{i=0}^{d-1} \binom{n-1}{i} \tag{5.41}$$

$$= \sum_{i=0}^{d} \left(\binom{n-1}{i} + \binom{n-1}{i-1} \right) = \sum_{i=0}^{d} \binom{n}{i}, \tag{5.42}$$

where the last equality uses Pascal's identity. Hence the claim holds for all n. ■

⊛ *The following proof is included out of personal interest and is not part of the course material. Readers who wish to focus solely on the course content may safely skip to the next part.*

Proof

Here we provide another proof.
Fix n and choose $S \subseteq \mathcal{X}$ with $|S| = n$ such that $m_{\mathcal{H}}(n) = |\mathcal{H}|_S|$, where

$$\mathcal{H}|_S := \{ (h(x))_{x \in S} : h \in \mathcal{H} \} \tag{5.43}$$

is the set of labelings realized on S. We prove by induction on n that

$$|\mathcal{H}|_S| \leq \sum_{i=0}^{d} \binom{n}{i}. \tag{5.44}$$

Base cases. *If $n = 0$, then $|\mathcal{H}|_S| = 1$ and*

$$\sum_{i=0}^{d} \binom{0}{i} = 1, \tag{5.45}$$

so the claim holds. If $d = 0$, then no single point can be labeled in two different ways, hence $|\mathcal{H}|_S| = 1$ for all n, while

$$\sum_{i=0}^{0} \binom{n}{i} = 1, \tag{5.46}$$

so the bound again holds.
Inductive step. *Assume the statement true for all pairs (n', d') with $n' < n$ and the same d. Pick some $x \in S$ and set*

$$S' := S \setminus \{x\}, \qquad |S'| = n - 1. \tag{5.47}$$

For each $h \in \mathcal{H}$, let $h|_{S'}$ be its restriction to S'. Define

$$\mathcal{A} := \{ h|_{S'} : h \in \mathcal{H} \}, \tag{5.48}$$

$$\mathcal{B} := \left\{ h|_{S'} : h \in \mathcal{H},\ \exists\, h' \in \mathcal{H} \text{ with } h'(x) \neq h(x),\ h'|_{S'} = h|_{S'} \right\}. \tag{5.49}$$

Thus, $\mathcal{A}$ is the set of all restrictions on S', and $\mathcal{B}$ collects those restrictions for which $\mathcal{H}$ contains at least two hypotheses that agree on S' but disagree on x.
Every labeling of S comes from choosing a restriction in $\mathcal{A}$, and then choosing the value at x. Labelings for which the value at x is forced *(no alternative hypothesis with the same restriction) contribute at most $|\mathcal{A}|$ patterns; labelings for which the value at x can be flipped (i.e. restrictions in $\mathcal{B}$) contribute at most $|\mathcal{B}|$ additional patterns. Hence*

$$|\mathcal{H}|_S| \leq |\mathcal{A}| + |\mathcal{B}|. \tag{5.50}$$

Note that $|\mathcal{A}| \leq m_{\mathcal{H}}(n-1)$, because $\mathcal{A}$ is a set of labelings on S' of size $n-1$. To bound $|\mathcal{B}|$, define a new hypothesis class

$$\mathcal{H}_x := \left\{ h|_{S'} : h \in \mathcal{H},\ \exists\, h' \in \mathcal{H} \text{ with } h'(x) \neq h(x),\ h'|_{S'} = h|_{S'} \right\}. \tag{5.51}$$

Then $\mathcal{B}$ is precisely $\mathcal{H}_x$ viewed as a set of labelings on S', so $|\mathcal{B}| \leq m_{\mathcal{H}_x}(n-1)$.

We claim that

$$\mathrm{dvc}(\mathcal{H}_x) \le d - 1. \tag{5.52}$$

Otherwise, suppose $\mathcal{H}_x$ shatters some $T \subseteq S'$ with $|T| = d$. By the definition of $\mathcal{H}_x$, for every labeling y_T on T and either choice of label $y_x \in \{-1, +1\}$ at x, there exists $h \in \mathcal{H}$ such that

$$h|_T = y_T, \qquad h(x) = y_x. \tag{5.53}$$

Hence $\mathcal{H}$ can realize all labelings on $T \cup \{x\}$, so $\mathcal{H}$ shatters a set of size $d + 1$, contradicting $\mathrm{dvc}(\mathcal{H}) = d$. Thus the claim holds.

By the induction hypothesis applied to $\mathcal{H}$ on $n - 1$ points and to $\mathcal{H}_x$ whose VC dimension is at most $d - 1$, we obtain

$$|\mathcal{A}| \le \sum_{i=0}^{d} \binom{n-1}{i}, \qquad |\mathcal{B}| \le \sum_{i=0}^{d-1} \binom{n-1}{i}. \tag{5.54}$$

Therefore,

$$|\mathcal{H}|_S| \le \sum_{i=0}^{d} \binom{n-1}{i} + \sum_{i=0}^{d-1} \binom{n-1}{i} \tag{5.55}$$

$$= \sum_{i=0}^{d} \left(\binom{n-1}{i} + \binom{n-1}{i-1} \right) \tag{5.56}$$

$$= \sum_{i=0}^{d} \binom{n}{i}. \tag{5.57}$$

■

Note

The inductive structure of the proof for the Sauer–Shelah Lemma is deeply reminiscent of strategies used in mathematical logic, *particularly in proving* completeness theorems *that connect syntactic derivability and semantic truth. In both settings, we start by imposing an ordering (or enumeration) on a collection of objects—whether they are formulas, sentences, or finite subsets of $\mathcal{X}$—so that induction can be applied coherently.*

Just as in logic we construct models step by step according to a fixed sequence of formulas (often by extending consistent sets in a well-ordered fashion), here we inductively analyze the growth of $\mathcal{H}$ by removing or reintroducing one element of the sample set at a time. The key similarity is structural: both arguments rely on well-founded recursion over a totally ordered domain to control how local extensions (adding one more formula or one more point) affect global consistency or combinatorial growth.

Remark

Plugging this bound into the standard uniform convergence inequality for a finite hypothesis class, we obtain a generalization bound of the form

$$\varepsilon = \sqrt{\frac{1}{2n} \log \frac{M}{\delta}} \sim O\left(\frac{\mathrm{dvc} \log n}{n} \right)^{1/2} \tag{5.58}$$

Thus, when $\mathrm{dvc}(\mathcal{H})$ is finite, the estimation error ε decreases as n grows, with the complexity of the class entering only through the polynomial factor controlled by d.

It can be rigorously shown that:

Theorem 5.5 (VC Generalization Bound)

With probability at least $1-\delta$*, the following inequality holds:*

$$E_{\text{out}}(h_i) - E_{\text{in}}(h_i) < \sqrt{\frac{8}{2n}\log\frac{4m_{\mathcal{H}}(2n)}{\delta}}. \tag{5.59}$$

Moreover,

$$E_{\text{out}}(h_i) - E_{\text{in}}(h_i) = O\left(\sqrt{\frac{\mathrm{d}_{\text{VC}}\log n}{n}}\right) \xrightarrow[n\to\infty]{} 0. \tag{5.60}$$

♡

Although the VC generalization bound is relatively loose in practice, its importance lies in converting an inherently infinite hypothesis space into a finitely bounded one. This theorem establishes a concrete analytical foundation for the entire field of machine learning, demonstrating that with sufficient data, generalization is not merely empirical, but theoretically guaranteed.